

Practical exercises — percentage change and simple interest

A Cambridge insert: increase, decrease, and the reverse question that catches everybody.

LESSON 131–132

2 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 10

LESSON CLOCK

14'

19'

23'

19'

5'

One multiplication, not two steps	14 min
Finding the change	19 min
Simple interest	23 min
The reverse question	19 min
Homework	5 min

LEARNING OBJECTIVES

- Increase and decrease a quantity by a given percentage in one multiplication.
- Find a percentage change from the old and new values.
- Calculate simple interest over several years.
- Find the original amount when the changed one is given.

TEXTBOOK

National	pp. 387–392
Cambridge	Stage 8, pp. 96–104
Workbook	Exercise 10.2

01

Explanation

One multiplication, not two steps

An increase of 15% multiplies by 1.15; a decrease of 15% multiplies by 0.85. Finding the part and then adding it is the same thing done twice.

new value = old value × $(1 \pm \frac{p}{100})$

CHANGE	MULTIPLIER	4000 BECOMES
up 15%	1.15	4600
down 15%	0.85	3400
up 8%	1.08	4320
down 40%	0.6	2400

UP 10% THEN DOWN 10% IS NOT BACK WHERE YOU STARTED

$100 \cdot 1.1 \cdot 0.9 = 99$. The two percentages are taken of different amounts, so they do not cancel — a fact every discount advertisement relies on.

Finding the change

$$\text{percentage change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100\%$$

OLD	NEW	CHANGE	PERCENTAGE
50	65	+15	+30%
80	60	−20	−25%
200	250	+50	+25%
250	200	−50	−20%

ALWAYS DIVIDE BY THE OLD VALUE

The last two rows are the same pair of numbers and give different percentages, because the base changed. A rise of 25% is undone by a fall of 20%, not of 25%.

Simple interest

$$I = \frac{P \cdot r \cdot t}{100}$$

with P the principal, r the yearly rate as a percentage and t the number of years.

Simple interest is calculated on the original sum every year.

P	R	T	INTEREST	TOTAL
1 000 000	12%	1	120 000	1 120 000
1 000 000	12%	3	360 000	1 360 000
500 000	8%	2	80 000	580 000

SIMPLE, NOT COMPOUND

Here the second year earns interest on 1 000 000 again, not on 1 120 000. Compound interest, where it does, waits until Grade 9 — but the difference is worth stating now.

The reverse question

“After a rise of 20% the price is 60 000. What was it before?” The 60 000 is **not** the base: divide by the multiplier rather than taking 20% of it.

GIVEN	WRONG	RIGHT	ANSWER
after +20% it is 60 000	$60\,000 - 12\,000$	$60\,000 \div 1.2$	50 000
after −20% it is 60 000	$60\,000 + 12\,000$	$60\,000 \div 0.8$	75 000
after +25% it is 500	$500 - 125$	$500 \div 1.25$	400

CHECK BY GOING FORWARDS

$50\,000 \cdot 1.2 = 60\,000$ ✓. The reverse question is the one most often got wrong in the whole of percentages, and the forward check takes five seconds.

02

Worked examples

EXAMPLE 1 A price of 4000 sum rises by 15%. Find the new price.

1

The multiplier is 1.15.

2

$4000 \cdot 1.15$

3

$= 4600$

ANSWER

4600 sum

EXAMPLE 2 1 000 000 sum is invested at 12% simple interest for 3 years. Find the interest and the total.

$$① \quad I = \frac{1\,000\,000 \cdot 12 \cdot 3}{100}$$

$$② \quad = 360\,000$$

Each year earns 120 000.

$$③ \quad \text{Total} = 1\,360\,000.$$

ANSWER

360 000 interest, 1 360 000 in all

EXAMPLE 3 After a rise of 20% a price is 60 000 sum. What was it before?

① The old price was multiplied by 1.2.

$$② \quad 60\,000 \div 1.2$$

Divide, do not subtract.

$$③ \quad = 50\,000$$

$$④ \quad \text{Check: } 50\,000 \cdot 1.2 = 60\,000 \checkmark$$

ANSWER

50 000 sum

03 Interactive model

Ask the class whether a 20% rise followed by a 20% fall returns the original price; the near-unanimous “yes” makes the arithmetic that follows memorable.

Multiplier, change, or reverse?

The multiplier for a rise of 15% is:

The multiplier for a fall of 15% is:

4000 up 15% is:

From 50 to 65 is a change of:

15%

23%

30%

130%

From 80 to 60 is a change of:

-20%

-25%

-33%

-75%

1 000 000 at 12% simple for 3 years earns:

120 000

360 000

404 928

36 000

After +20% a price is 60 000; before it was:

48 000

50 000

72 000

40 000

Up 10% then down 10% gives:

the same value

99% of it

101% of it

90% of it

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Percentage	Foiz	Процент
Increase	Ortish	Увеличение
Decrease	Kamayish	Уменьшение
Multiplier	Ko'paytuvchi	Множитель
Percentage change	Foizli o'zgarish	Процентное изменение
Simple interest	Oddiy foiz	Простые проценты
Principal	Asosiy summa	Основная сумма
Original value	Boshlang'ich qiymat	Первоначальное значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To increase by 8% multiply by:

To decrease by 40% multiply by:

Percentage change divides by:

Simple interest for t years is:

$$\frac{Prt}{100}$$

$$\frac{Pr}{100}$$

$$P(1 + r)^t$$

$$Prt$$

After -20% a price is 60 000; before it was:

72 000

75 000

48 000

80 000

A rise of 25% is undone by a fall of:

25%

20%

30%

75%

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 4000 up 15%

ANSWER 4600

2. 4000 down 15%

ANSWER 3400

3. 4000 up 8%

ANSWER 4320

4. 4000 down 40%

ANSWER 2400

5. The multiplier for +12%

ANSWER 1.12

6. The multiplier for -7%

ANSWER 0.93

7. 20% of 350

ANSWER 70

Medium · the standard question

7 PROBLEMS

1. From 50 to 65

ANSWER +30%

2. From 80 to 60

ANSWER -25%

3. 1 000 000 at 12% simple, 3 years: the interest

ANSWER 360 000

4. 500 000 at 8% simple, 2 years: the total

ANSWER 580 000

5. After +20% it is 60 000: before

ANSWER 50 000

6. After -20% it is 60 000: before

ANSWER 75 000

7. Up 10% then down 10% from 100

ANSWER 99

Hard · stretch and reason

7 PROBLEMS

1. A rise of 25% is undone by a fall of

ANSWER 20%

2. After two rises of 10% from 200

ANSWER 242

3. A price falls 30% to 49 000: the original

ANSWER 70 000

4. At 10% simple, how long until P doubles?

ANSWER 10 years

5. P at 15% simple for 4 years: the total as a multiple of P

ANSWER $1.6P$

6. A shop raises a price 20% then advertises 20% off: the net change

ANSWER -4%

7. Interest of 90 000 on 750 000 over 2 years: the rate

ANSWER 6%

07 Homework

Homework — 5 tasks

For every reverse question, check your answer by going forwards.

1. Increase 25 000 by 18% and decrease it by 18%.
2. A price rises from 40 000 to 46 000. Find the percentage change.
3. Find the simple interest on 2 000 000 at 9% for 4 years.
4. After a fall of 25% a price is 90 000. What was it before?
5. Show that a rise of 50% followed by a fall of 50% loses a quarter of the original.

Construction problems with compasses and a straightedge

Four basic constructions combined into triangles — and the rule that decides when it can be done at all.

LESSON 133–134

2 × 40 MIN

MATEMATIKA 7, §58

S8 5.3

LESSON CLOCK

9'

23'

24'

19'

5'

The rules of the game	9 min
A triangle from three sides	23 min
The other two triangles	24 min
Perpendiculars and midpoints	19 min
Homework	5 min

LEARNING OBJECTIVES

- Construct a triangle from three sides, from two sides and the angle between, and from a side and two angles.
- Decide when a construction is impossible.
- Construct the perpendicular from a point to a line and the midpoint of a segment.
- Justify a construction with a congruence criterion.

TEXTBOOK

National	pp. 393–400
Cambridge	Stage 8, pp. 52–56
Workbook	Exercise 5.3

01

Explanation

The rules of the game

Only two tools are allowed: a straightedge, which draws a line through two points, and compasses, which draw a circle of a given radius. No measuring with the ruler's scale, and no protractor.

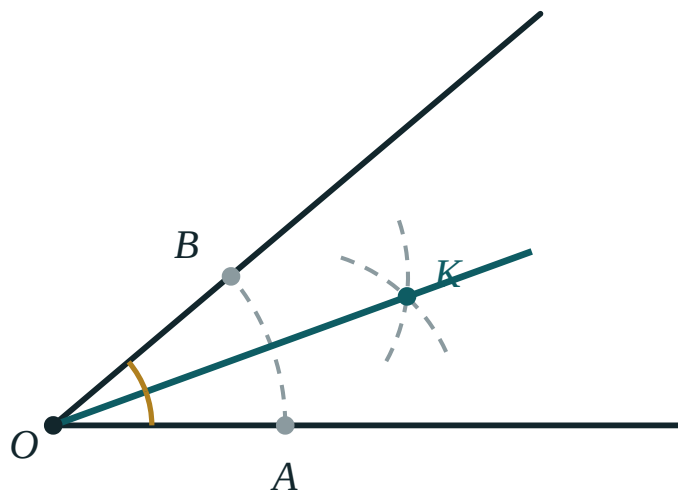
ALLOWED	NOT ALLOWED
a line through two points	measuring a length with the scale
a circle with a given centre and radius	measuring an angle with a protractor
marking where two drawn curves cross	sliding the ruler until it looks right

LEAVE EVERY ARC ON THE PAGE

The arcs are the proof that the construction was done and not estimated. Rubbing them out removes the working, and with it the marks.

A triangle from three sides

Given three lengths a , b , c :



Arcs meeting to fix the third vertex

STEP	WHAT TO DO
1	draw a segment $BC = a$
2	with centre B and radius c , draw an arc
3	with centre C and radius b , draw a second arc
4	call the crossing A and join AB and AC

Every triangle built this way is congruent to every other, by the third criterion — which is why the three sides determine the triangle completely.

THE ARCS MEET ONLY IF THE TRIANGLE INEQUALITY HOLDS

With 5, 8, 14 the arcs fall short of each other and there is nothing to mark. The construction fails exactly when the triangle does not exist.

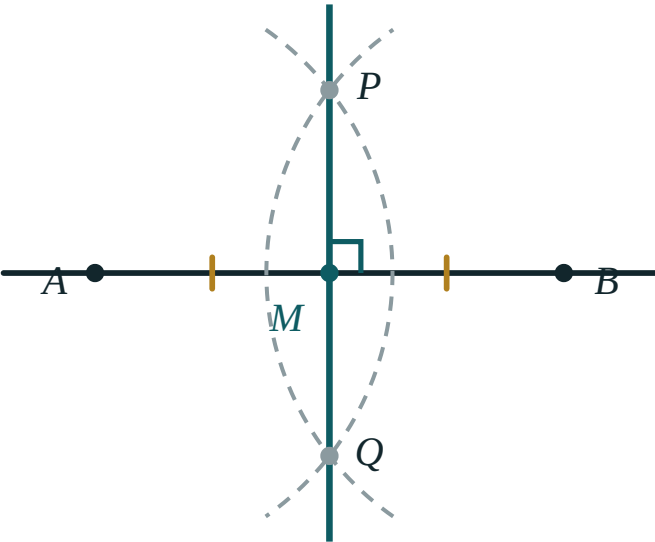
The other two triangles

GIVEN	CONSTRUCTION	JUSTIFIED BY
three sides	two arcs from the ends of the first side	the third criterion
two sides and the angle between	copy the angle, then mark the two lengths along its arms	the first criterion
a side and the two angles on it	copy each angle at one end of the side	the second criterion

THE THREE CRITERIA ARE THE THREE CONSTRUCTIONS

Each congruence criterion says which data fix a triangle; each construction is the recipe for building the triangle those data describe. The chapters answer each other.

Perpendiculars and midpoints



The perpendicular bisector of a segment

CONSTRUCTION	METHOD	WHAT IT ALSO GIVES
perpendicular bisector of AB	equal arcs from A and from B , joined	the midpoint of AB
perpendicular from P to a line	an arc from P cutting the line twice, then the bisector of that segment	the distance from P to the line
bisector of an angle	equal arcs from the vertex, then from the two crossings	the locus of equidistant points
an angle of 60°	an equilateral triangle on a segment	30°

WHY THE PERPENDICULAR BISECTOR WORKS

Both arc centres are the same distance from the two crossing points, so each crossing is equidistant from A and B . The line through them is the set of all such points — and that line meets AB at right angles, at its midpoint.

02 Worked examples

EXAMPLE 1 Construct a triangle with sides 6 cm, 5 cm and 4 cm.

- 1

Check: $4 + 5 > 6$ ✓
The longest side only.
- 2

Draw $BC = 6$ cm.
- 3

Arc of 5 cm from B , arc of 4 cm from C .
- 4

Join the crossing to both ends.
Congruent by SSS to any other such triangle.

ANSWER

Constructed

EXAMPLE 2 Can a triangle with sides 5, 8, 14 be constructed?

- ① The longest side is 14.
- ② $5 + 8 = 13 < 14$
- ③ The two arcs never meet.
- ④ No such triangle exists.

ANSWER

No

EXAMPLE 3 Construct the midpoint of a segment AB .

- ① With centre A and radius more than half AB , draw an arc.
- ② Repeat from B with the same radius.
- ③ Join the two crossings.
- ④ That line meets AB at its midpoint, at right angles.

ANSWER

The perpendicular bisector gives it

03 Interactive model

Set 5, 8 and 14 cm without comment; the class works for two minutes before discovering the arcs cannot meet, and the triangle inequality is never forgotten again.

Which construction, and does it work?

The two allowed tools are:

ruler and protractor

straightedge and compasses

compasses and protractor

ruler and set square

Arcs should be:

rubbed out

left on the page

drawn in pen

omitted

Three sides fix a triangle by:

the first criterion

the second

the third

none

Sides 5, 8, 14 can be constructed:

yes

no

only with a protractor

sometimes

Two sides and the angle between them use:

the first criterion

the second

the third

none

The perpendicular bisector also gives:

the midpoint

the bisector of an angle

the area

the centroid

An equilateral triangle constructs an angle of:

45°

60°

90°

30° directly

Every point of the perpendicular bisector is:

on the segment

equidistant from the ends

the midpoint

outside

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Construction	Yasash	Построение
Compasses	Sirkul	Циркуль
Straightedge	Chizg'ich	Линейка
Arc	Yoy	Дуга
Perpendicular bisector	O'rta perpendikulyar	Серединный перпендикуляр
Midpoint	O'rta nuqta	Середина
Given data	Berilganlar	Данные
Impossible	Yasab bo'lmaydi	Невозможно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Compasses may be used to:

measure an angle

draw a circle of given radius

draw a straight line

estimate

The construction from three sides fails when:

the sides are equal

the triangle inequality fails

one side is long

never

A side and the two angles on it use:

the first criterion

the second criterion

the third criterion

none

The perpendicular bisector of AB passes through:

A

the midpoint of AB

B

no special point

Construction arcs should be:

erased

left visible

coloured

redrawn

A construction is justified by:

measuring the result

a congruence criterion

the drawing looking right

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The two allowed tools

ANSWER straightedge and compasses

2. Three sides fix a triangle by

ANSWER the third criterion

3. Two sides and the included angle

ANSWER the first criterion

4. A side and two angles on it

ANSWER the second criterion

5. Sides 6, 5, 4: constructible?

ANSWER Yes

6. Sides 5, 8, 14: constructible?

ANSWER No

7. The perpendicular bisector gives the

ANSWER midpoint

Medium · the standard question

7 PROBLEMS

1. Sides 3, 4, 8: constructible?

ANSWER No

2. Sides 7, 7, 13: constructible?

ANSWER Yes

3. Which construction gives an angle of 60° ?

ANSWER An equilateral triangle

4. And of 30° ?

ANSWER Bisect the 60°

5. And of 90° ?

ANSWER A perpendicular to a line at a point

6. And of 45° ?

ANSWER Bisect the 90°

7. Why leave the arcs?

ANSWER They show the construction was not estimated

Hard · stretch and reason

7 PROBLEMS

1. Construct a triangle from two sides 5 and 7 with the angle between them 60°

ANSWER Copy the angle, mark the lengths on its arms

2. How many triangles arise from three given sides?

ANSWER One, up to congruence

3. Construct an angle of 15°

ANSWER Bisect 60° twice

4. Construct the perpendicular from a point to a line

ANSWER An arc cutting the line twice, then bisect that segment

5. Why does the perpendicular-bisector construction work?

ANSWER Both crossings are equidistant from the two ends

6. Given the perimeter and two angles, is the triangle fixed?

ANSWER Yes — the third angle follows and the shape scales to the perimeter

7. Sides 2, 3, 5: constructible?

ANSWER No — the points would be in a line

07 Homework

Homework — 5 tasks

Check the triangle inequality before touching the compasses.

1. Construct a triangle with sides 7 cm, 5 cm and 4 cm.
2. Try to construct one with sides 3 cm, 4 cm and 9 cm, and explain what happens.
3. Construct an angle of 60° and bisect it.
4. Construct the perpendicular bisector of a 8 cm segment and mark its midpoint.
5. Name the congruence criterion that justifies each of the three triangle constructions.

Practical exercises — the circumference of a circle

A Cambridge insert: why $C \div d$ is the same number for every circle in the world.

LESSON 135–136

2 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 8.2

LESSON CLOCK

18'	18'	18'	22'	4'
The measuring experiment		18 min		
The two formulae		18 min		
Backwards		18 min		
Wheels and tracks		22 min		
Homework		4 min		

LEARNING OBJECTIVES

- Measure C and d for several circles and find their ratio.
- Use $C = \pi d$ and $C = 2\pi r$ in both directions.
- Find the area of a circle with $S = \pi r^2$.
- Solve practical problems about wheels, tracks and circular edges.

TEXTBOOK

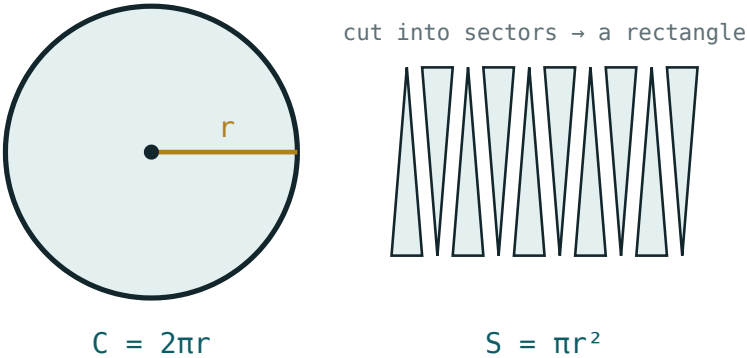
National	pp. 401–406
Cambridge	Stage 8, pp. 82–88
Workbook	Exercise 8.2

01

Explanation

The measuring experiment

Wrap a thread round several round objects, measure the length and the diameter, and divide one by the other.



A circle, its radius and its diameter

OBJECT	D	C	$C \div D$
a coin	2.4 cm	7.5 cm	3.13
a cup	8 cm	25.2 cm	3.15
a bucket	28 cm	88 cm	3.14
a wheel	60 cm	188 cm	3.13

THE SAME NUMBER EVERY TIME

Whatever the size, the ratio lands near 3.14 . That constant is called π , and the small differences in the table are measuring error, not a difference between the circles.

The two formulae

$C = \pi d = 2\pi r$ $S = \pi r^2$

R	D	C	S
5 cm	10 cm	31.4 cm	78.5 cm^2
7 cm	14 cm	44 cm	154 cm^2
10 cm	20 cm	62.8 cm	314 cm^2

C USES R ONCE, S USES IT TWICE

Doubling the radius doubles the circumference but multiplies the area by 4. Using πd for an area, or πr^2 for a length, is the standard confusion — check the units: cm for a length, cm^2 for an area.

Backwards

GIVEN	WORKING	ANSWER
$C = 62.8\text{ cm}$	$d = C \div \pi$	$d = 20\text{ cm}$
$C = 44\text{ cm}$	$r = C \div (2\pi)$	$r = 7\text{ cm}$
$S = 78.5\text{ cm}^2$	$r^2 = S \div \pi$	$r = 5\text{ cm}$
$S = 314\text{ cm}^2$	$r^2 = 100$	$r = 10\text{ cm}$

TWO USEFUL APPROXIMATIONS

$\pi \approx 3.14$ suits a calculator; $\pi \approx \frac{22}{7}$ suits mental work whenever the radius or diameter is a multiple of 7 — which is why textbook circles so often have radius 7 or 14.

Wheels and tracks

PROBLEM	IDEA	ANSWER
a wheel of diameter 70 cm : distance in one turn	$C = \pi d$	$\approx 220\text{ cm}$
how many turns in 220 m	$22000 \div 220$	100
a circular pond of radius 14 m : the fence	$C = 2\pi r$	$\approx 88\text{ m}$
and the water surface	$S = \pi r^2$	$\approx 616\text{ m}^2$
a running track of radius 35 m : one lap	$C = 2\pi r$	$\approx 220\text{ m}$

WHERE π CAME FROM

Al-Kashi, working in Samarkand in 1424, computed π to sixteen decimal places — a record that stood for nearly two centuries. The constant measured with thread in this lesson has a long local history.

02 Worked examples

EXAMPLE 1 A circle has radius 7 cm . Find its circumference and area, taking $\pi = \frac{22}{7}$.

$$① \quad C = 2\pi r = 2 \cdot \frac{22}{7} \cdot 7$$

$$② \quad C = 44\text{ cm}$$

$$③ \quad S = \pi r^2 = \frac{22}{7} \cdot 49$$

$$④ \quad S = 154\text{ cm}^2$$

ANSWER

44 cm and 154 cm^2

EXAMPLE 2 A circle has circumference 62.8 cm . Find its radius, taking $\pi = 3.14$.

$$① \quad C = 2\pi r$$

$$② \quad 62.8 = 6.28r$$

$$③ \quad r = 10\text{ cm}$$

ANSWER

10 cm

EXAMPLE 3 A wheel of diameter 70 cm rolls 220 m . How many turns does it make?

$$\textcircled{1} \quad C = \pi d = \frac{22}{7} \cdot 70 = 220\text{ cm}$$

One turn.

$$\textcircled{2} \quad 220\text{ m} = 22\,000\text{ cm}$$

Same units.

$$\textcircled{3} \quad 22\,000 \div 220 = 100$$

ANSWER

100 turns

03 Interactive model

Do the thread measurement first and tabulate every group's ratio on the board; the class discovers π rather than being handed it.

Length or area?

$C \div d$ equals:

The circumference of a circle is:

The area of a circle is:

$r = 7$, $\pi = \underline{22}$ 7 : the circumference is:

22

44

154

49

The same circle has area:

44

154

308

49

Doubling the radius multiplies the area by:

2

3

4

8

$C = 62.8$ with $\pi = 3.14$ gives r equal to:

5

10

20

31.4

An answer in cm^2 must be:

a circumference

an area

a radius

a diameter

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Circle	Aylana	Окружность
Circumference	Aylana uzunligi	Длина окружности
Diameter	Diametr	Диаметр
Radius	Radius	Радиус
Pi	Pi soni	Число пи
Ratio	Nisbat	Отношение
Disc	Doira	Круг
Revolution	Aylanish	Оборот

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

π is the ratio of:

C to r

C to d

S to r

d to r

π is about:

3.41

3.14

2.14

1.34

A circle of radius 5 has circumference about:

15.7

31.4

78.5

25

And area about:

31.4

78.5

157

25

A wheel of diameter 70 cm covers per turn about:

70 cm

140 cm

220 cm

440 cm

$S = 314\text{ cm}^2$ gives r equal to:

5

10

20

100

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 5$: the circumference ($\pi = 3.14$)

ANSWER 31.4 cm

2. $r = 5$: the area

ANSWER 78.5 cm^2

3. $r = 7$: the circumference ($\pi = \frac{22}{7}$)

ANSWER 44 cm

4. $r = 7$: the area

ANSWER 154 cm^2

5. $d = 20$: the circumference

ANSWER 62.8 cm

6. $r = 10$: the area

ANSWER 314 cm^2

7. $C \div d$

ANSWER π

Medium · the standard question

7 PROBLEMS

1. $C = 62.8$: the radius

ANSWER 10 cm

2. $C = 44$: the radius

ANSWER 7 cm

3. $S = 78.5$: the radius

ANSWER 5 cm

4. A wheel of diameter 70 cm : one turn

ANSWER 220 cm

5. How many such turns in 220 m ?

ANSWER 100

6. A pond of radius 14 m : the fence

ANSWER 88 m

7. And the surface

ANSWER 616 m^2

Hard · stretch and reason

7 PROBLEMS

1. Doubling the radius multiplies the area by

ANSWER 4

2. A track of radius 35 m: one lap

ANSWER 220 m

3. How many laps make 1100 m?

ANSWER 5

4. A circle of area 154 cm²: its circumference

ANSWER 44 cm

5. The area of a semicircle of radius 7

ANSWER 77 cm²

6. A ring between circles of radii 7 and 14: its area

ANSWER 462 cm²

7. Who computed π to sixteen places in Samarkand?

ANSWER Al-Kashi, in 1424

07 Homework

Homework — 5 tasks

Write the units with every answer: *cm* for a length, *cm²* for an area.

- Find the circumference and area of a circle of radius 21 cm, taking $\pi = \frac{22}{7}$.
- A circle has circumference 88 cm. Find its radius and area.
- A wheel of diameter 56 cm makes 50 turns. How far does it travel?
- Find the area of a ring between circles of radii 10 cm and 6 cm.

5. Measure three round objects at home and tabulate $C \div d$ for each.

The Cartesian coordinate system

Two number lines at right angles, and every point of the plane named by an ordered pair.

LESSON 137–138

2 × 40 MIN

MATEMATIKA 7, §59

S8 11

LESSON CLOCK

14'

19'

23'

19'

5'

Two axes and one origin	14 min
Plotting and reading	19 min
The four quadrants	23 min
Lengths and midpoints	19 min
Homework	5 min

LEARNING OBJECTIVES

- Plot a point from its coordinates and read the coordinates of a plotted point.
- Name the four quadrants and the signs in each.
- Recognise the points of the axes by their coordinates.
- Find lengths and midpoints of segments parallel to an axis.

TEXTBOOK

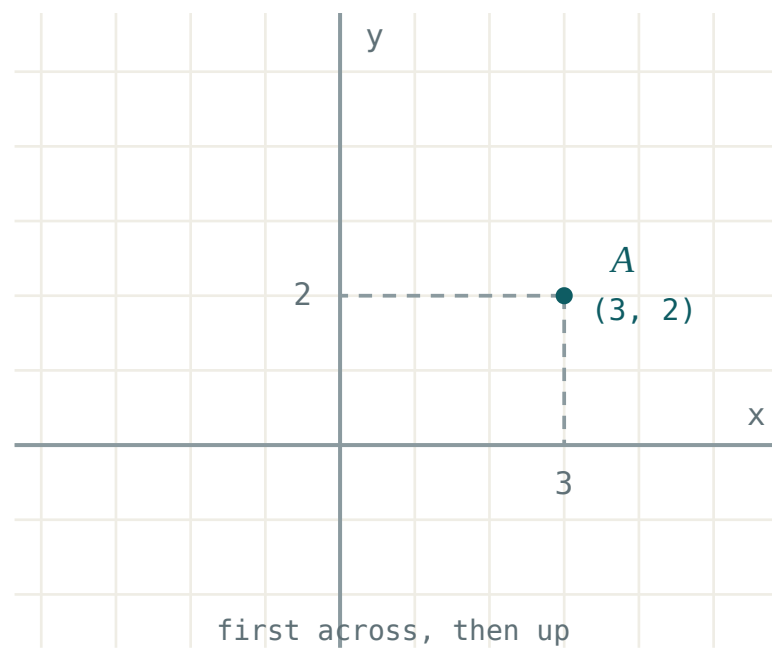
National	pp. 407–414
Cambridge	Stage 8, pp. 106–116
Workbook	Exercise 11.1

01

Explanation

Two axes and one origin

Two perpendicular number lines meet at the **origin** $O(0; 0)$: the horizontal Ox and the vertical Oy . Every point is then named by an ordered pair.



A point and its two coordinates

NAME	WHICH NUMBER	READ FROM
abscissa x	the first	the horizontal axis
ordinate y	the second	the vertical axis

THE ORDER IS FIXED: ALONG FIRST, THEN UP

$A(3; 2)$ and $B(2; 3)$ are different points. “Ordered pair” means exactly this — the pair carries information in its order, not only in its two numbers.

Plotting and reading

POINT	GO ALONG	THEN	WHERE IT LANDS
$A(3; 2)$	3	up 2	quadrant I
$B(-4; 1)$	left 4	up 1	quadrant II
$C(-2; -3)$	left 2	down 3	quadrant III
$D(5; -1)$	5	down 1	quadrant IV
$E(0; 4)$	nowhere	up 4	on the y -axis
$F(6; 0)$	6	nowhere	on the x -axis

A ZERO TELLS YOU WHICH AXIS

$x = 0$ puts the point on the vertical axis; $y = 0$ puts it on the horizontal one. Both zero is the origin itself.

The four quadrants

QUADRANT	WHERE	X	Y	EXAMPLE
I	top right	+	+	(3; 2)
II	top left	−	+	(−4; 1)
III	bottom left	−	−	(−2; −3)
IV	bottom right	+	−	(5; −1)

THEY ARE NUMBERED ANTICLOCKWISE

Starting at the top right and turning the way a positive angle turns. The numbering is a convention, but it is the same convention worldwide.

Lengths and midpoints

When a segment is parallel to an axis, its length is the difference of the coordinates that change, and its midpoint is their average.

SEGMENT	LENGTH	MIDPOINT
A(1; 3) to B(7; 3)	6	(4; 3)
C(2; −1) to D(2; 5)	6	(2; 2)
E(−3; 4) to F(4; 4)	7	(0.5; 4)

THE IDEA BEHIND THE COORDINATE PLANE

Descartes' insight was that a geometric question — where is the midpoint? — becomes an arithmetic one: take the average. Grade 8 turns every line and circle into an equation on exactly this principle.

02 Worked examples**EXAMPLE 1** In which quadrant does $(-2; -3)$ lie?

- ① x is negative: left of the y -axis.
- ② y is negative: below the x -axis.
- ③ Bottom left — quadrant III.

ANSWER

Quadrant III

EXAMPLE 2 Find the length and the midpoint of the segment from $A(1; 3)$ to $B(7; 3)$.

- ① The ordinates are equal, so the segment is horizontal.
- ② Length = $7 - 1 = 6$.
- ③ Midpoint abscissa = $\frac{1 + 7}{2} = 4$.
- ④ Midpoint $(4; 3)$.

ANSWERLength 6, midpoint $(4; 3)$

EXAMPLE 3 Three vertices of a rectangle are $(1; 1)$, $(5; 1)$ and $(5; 4)$. Find the fourth.

1 The side from $(1; 1)$ to $(5; 1)$ is horizontal.

2 The side from $(5; 1)$ to $(5; 4)$ is vertical.

3 The fourth vertex is above $(1; 1)$ by 3.

4 $(1; 4)$

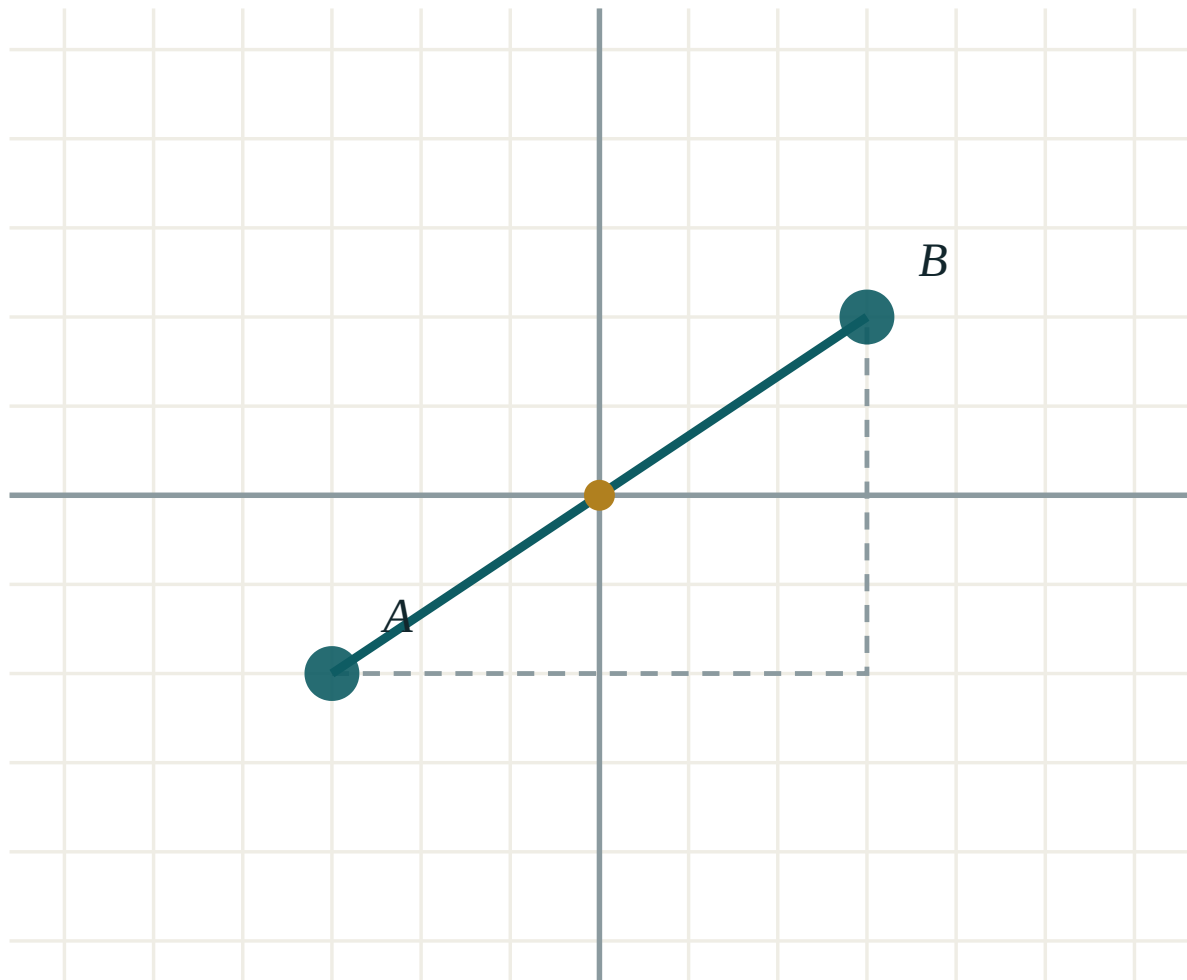
ANSWER

$(1; 4)$

03 Interactive model

Play “battleships” on squared paper for ten minutes with coordinates instead of letters; the order of the pair stops being a rule and becomes a habit.

Two points, their difference and their midpoint



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Coordinate plane	Koordinata tekisligi	Координатная плоскость
Axis	O'q	Ось
Origin	Koordinata boshi	Начало координат
Abscissa	Abssissa	Абсцисса
Ordinate	Ordinata	Ордината
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
Quadrant	Chorak	Четверть
To plot	Belgilamoq	Отметить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first number of a pair is the:

ordinate

abscissa

origin

quadrant

$(3; 2)$ and $(2; 3)$ are:

the same point

different points

both on an axis

both in quadrant II

$(-4; 1)$ lies in quadrant:

I

II

III

IV

$(5; -1)$ lies in quadrant:

I

II

III

IV

$(0; 4)$ lies:

in quadrant I

on the y -axis

on the x -axis

at the origin

The midpoint of $(1; 3)$ and $(7; 3)$ is:

$(4; 3)$

$(3; 3)$

$(8; 6)$

$(4; 6)$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The quadrant of $(3; 2)$

ANSWER I

2. The quadrant of $(-4; 1)$

ANSWER II

3. The quadrant of $(-2; -3)$

ANSWER III

4. The quadrant of $(5; -1)$

ANSWER IV

5. Where is $(0; 4)$?

ANSWER On the y-axis

6. Where is $(6; 0)$?

ANSWER On the x-axis

7. The coordinates of the origin

ANSWER $(0; 0)$

Medium · the standard question

7 PROBLEMS

1. Length of $A(1; 3)$ to $B(7; 3)$

ANSWER 6

2. Its midpoint

ANSWER $(4; 3)$

3. Length of $C(2; -1)$ to $D(2; 5)$

ANSWER 6

4. Its midpoint

ANSWER $(2; 2)$

5. The fourth vertex of a rectangle on $(1; 1)$, $(5; 1)$, $(5; 4)$

ANSWER $(1; 4)$

6. The reflection of $(3; 2)$ in the x -axis

ANSWER $(3; -2)$

7. The reflection of $(3; 2)$ in the y -axis

ANSWER $(-3; 2)$

Hard · stretch and reason

7 PROBLEMS

1. The area of the rectangle with vertices $(1; 1)$, $(5; 1)$, $(5; 4)$, $(1; 4)$

ANSWER 12

2. Its perimeter

ANSWER 14

3. All points with $y = 3$ form

ANSWER a horizontal line

4. All points with $x = -2$ form

ANSWER a vertical line

5. All points with $x = y$ form

ANSWER the line through the origin bisecting quadrants I and III

6. The midpoint of $(-3; 4)$ and $(4; 4)$

ANSWER $(0.5; 4)$

7. A point equidistant from both axes in quadrant III

ANSWER e.g. $(-2; -2)$

07 Homework

Homework — 5 tasks

Along first, then up — and label every point you plot.

- Plot $A(2; 5)$, $B(-3; 4)$, $C(-1; -2)$ and $D(4; -3)$, and name each quadrant.
- Find the length and midpoint of the segment from $(-2; 3)$ to $(6; 3)$.
- Three vertices of a rectangle are $(2; 1)$, $(8; 1)$ and $(8; 5)$. Find the fourth and the area.
- Where do all the points with $y = -4$ lie?
- Reflect $(-5; 2)$ in each axis and give both images.

The idea of a function

A rule that gives each input exactly one output — as a table, a formula, and a graph.

LESSON 139–140

2 × 40 MIN

MATEMATIKA 7, §60

S8 9.4

LESSON CLOCK

14'

19'

23'

19'

5'

What a function is	14 min
Three ways of giving one	19 min
Values, forwards and backwards	23 min
The domain	19 min
Homework	5 min

LEARNING OBJECTIVES

- State what makes a rule a function.
- Move between a table, a formula and a graph of the same function.
- Find the value of a function at a given argument, and the argument for a given value.
- Name the domain of a simple function.

TEXTBOOK

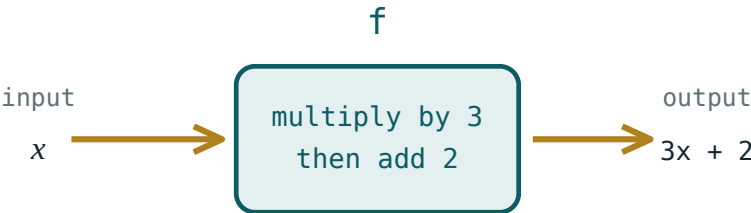
National	pp. 415–422
Cambridge	Stage 8, pp. 90–95
Workbook	Exercise 9.4

01

Explanation

What a function is

A **function** is a rule that assigns to each value of x exactly one value of y . One input, one output — never two.



A function as a machine: one input, one output

RULE	A FUNCTION?	WHY
y is twice x	yes	one answer every time
y is the area of a square of side x	yes	$y = x^2$
y is a number whose square is x	no	$x = 9$
y is the day of the week for a date x	yes	each date has one day

THE VERTICAL-LINE TEST

On a graph, a rule is a function exactly when no vertical line meets the curve twice — because a second meeting would be a second y for the same x .

Three ways of giving one

WAY	EXAMPLE	BEST FOR
a formula	$y = 2x + 1$	calculating any value exactly
a table	a row of x and a row of y	seeing a few values at once
a graph	points $(x; y)$ plotted	seeing the shape of the change
words	“double it and add one”	stating the rule to somebody

x	-2	-1	0	1	2	3
$y = 2x + 1$	-3	-1	1	3	5	7
$y = x^2$	4	1	0	1	4	9

ALL THREE DESCRIBE ONE OBJECT

The table is a sample of the formula; the graph is the whole table drawn at once. Being able to move between them is the skill this lesson is really about.

Values, forwards and backwards

Write $f(x)$ for the value at x . Finding $f(3)$ is substitution; finding x from a known $f(x)$ is solving an equation.

QUESTION	WORKING	ANSWER
$f(x) = 2x + 1, f(3)$	$2 \cdot 3 + 1$	7
$f(x) = 2x + 1, f(-4)$	$-8 + 1$	-7
$f(x) = 2x + 1 = 11$	$2x = 10$	$x = 5$
$f(x) = x^2, f(x) = 25$	$x^2 = 25$	$x = \pm 5$

ONE Y MAY COME FROM TWO XS

A function forbids two outputs for one input, not two inputs for one output. $f(x) = x^2$ sends both 5 and -5 to 25 and is still perfectly a function.

The domain

The **domain** is the set of values of x for which the rule gives an answer.

FUNCTION	DOMAIN	WHY
$y = 2x + 1$	every number	nothing can go wrong
$y = \frac{1}{x}$	$x \neq 0$	division by zero
$y = \frac{1}{x - 3}$	$x \neq 3$	the denominator vanishes
the area of a square of side x	$x > 0$	a side cannot be negative

TWO KINDS OF RESTRICTION

One comes from the algebra — a zero denominator; the other from the situation — a length must be positive. Both are part of the answer when a function describes something real.

02 Worked examples**EXAMPLE 1** For $f(x) = 2x + 1$, find $f(3)$ and $f(-4)$.

① $f(3) = 2 \cdot 3 + 1 = 7$

② $f(-4) = 2 \cdot (-4) + 1$

③ $= -8 + 1 = -7$

ANSWER

7 and -7

EXAMPLE 2 For $f(x) = 2x + 1$, find x when $f(x) = 11$.

① $2x + 1 = 11$
Now it is an equation.

② $2x = 10$

③ $x = 5$

ANSWER $x = 5$

EXAMPLE 3 Is “ y is a number whose square is x ” a function?

- ① Take $x = 9$.
- ② Both 3 and -3 qualify.
- ③ One input gives two outputs.
- ④ So it is not a function.

ANSWER

No

03 Interactive model

Draw a function machine on the board and feed it numbers the class calls out; the definition arrives as a description of what everyone has just watched.

Feed a value into $\frac{1}{x-3}$

$$f(x) = \frac{1}{x-3}$$

x 5

denominator 2

value 0.5

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Function	Funksiya	Функция
Argument	Argument	Аргумент
Value of a function	Funksiya qiymati	Значение функции
Independent variable	Erkli o'zgaruvchi	Независимая переменная
Dependent variable	Bog'liq o'zgaruvchi	Зависимая переменная
Domain	Aniqlanish sohasi	Область определения
Table of values	Qiymatlar jadvali	Таблица значений
Graph	Grafik	График

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A function gives each input:

The vertical-line test fails when a vertical line:

For $f(x) = 2x + 1$, $f(3)$ is:

$f(x) = 11$ gives x equal to:

The domain of $y = \underline{1}x - 3$ is:

$f(x) = x^2$ sends both 5 and -5 to 25, so it is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $f(x) = 2x + 1: f(0)$

ANSWER 1

2. $f(3)$

ANSWER 7

3. $f(-4)$

ANSWER -7

4. $f(x) = x^2: f(3)$

ANSWER 9

5. $f(-3)$

ANSWER 9

6. The domain of $y = 2x + 1$

ANSWER Every number

7. The domain of $y = \frac{1}{x}$

ANSWER $x \neq 0$

Medium · the standard question

7 PROBLEMS

1. $f(x) = 2x + 1 = 11$: find x

ANSWER $x = 5$

2. $f(x) = x^2 = 25$: find x

ANSWER $x = \pm 5$

3. The domain of $y = \frac{1}{x-3}$

ANSWER $x \neq 3$

4. Is " y^2 equals x " a function of x ?

ANSWER No

5. Is the area of a square a function of its side?

ANSWER Yes

6. A table for $y = 2x + 1$ at $x = -2, 0, 2$

ANSWER $-3, 1, 5$

7. The vertical-line test decides

ANSWER whether a graph shows a function

Hard · stretch and reason

7 PROBLEMS

1. $f(x) = 3x - 2$: find $f(f(1))$

ANSWER 1

2. $f(x) = x^2 - 4$: find every x with $f(x) = 0$

ANSWER $x = \pm 2$

3. The domain of $y = \frac{1}{x^2 - 4}$

ANSWER $x \neq \pm 2$

4. A function whose graph is a horizontal line

ANSWER $y = c$

5. Why is a vertical line not the graph of a function?

ANSWER One x would have every y

6. $f(x) = 2x + 1$: for which x is $f(x) = x$?

ANSWER $x = -1$

7. The domain of the perimeter of a square as a function of its side

ANSWER $x > 0$

07 Homework

Homework — 5 tasks

Say which is the argument and which is the value in every task.

- For $f(x) = 3x - 5$, find $f(0)$, $f(2)$ and $f(-1)$.
- For the same function, find x when $f(x) = 7$.
- Make a table of $y = x^2 - 1$ for $x = -3$ to 3 .
- Give the domain of $y = \frac{1}{x + 2}$.

5. Explain why “ y is a number whose square is x ” is not a function.

The linear function

$y = kx + b$ — a straight line, with k its steepness and b where it crosses.

LESSON 141–142

2 × 40 MIN

MATEMATIKA 7, §61

S8 9.2–9.3

LESSON CLOCK

14'

23'

19'

19'

5'

Two points are enough	14 min
What k and b do	23 min
Meeting the axes	19 min
Is the point on the line?	19 min
Homework	5 min

LEARNING OBJECTIVES

- Draw the graph of $y = kx + b$ from two points.
- Interpret k as the gradient and b as the intercept.
- Find where the line meets each axis.
- Decide whether a point lies on a given line.

TEXTBOOK

National	pp. 423–431
Cambridge	Stage 8, pp. 88–95
Workbook	Exercise 9.3

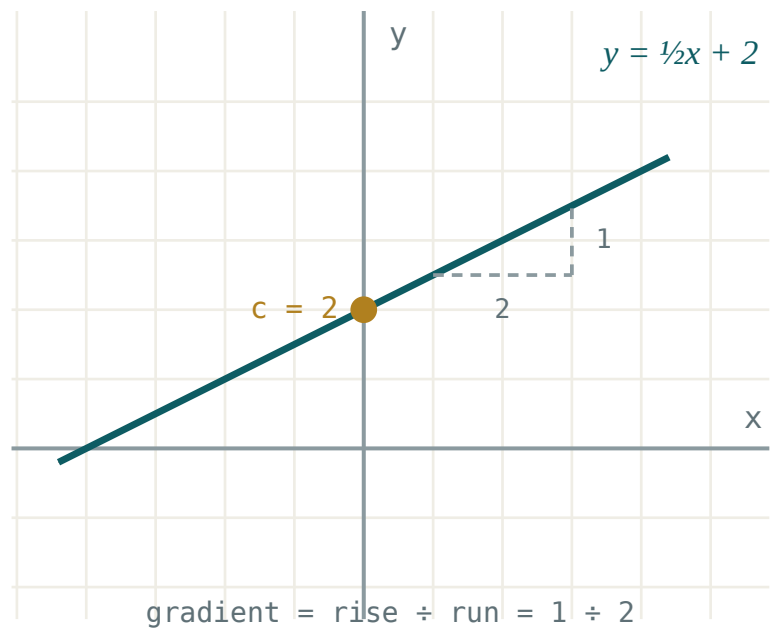
01

Explanation

Two points are enough

$$y = kx + b$$

Every such function has a straight line for its graph, and two points determine a straight line — so a table of two values is all that is needed. A third point is worth plotting as a check.



The gradient and the intercept of a line

FUNCTION	$X = 0$	$X = 1$	LINE THROUGH
$y = 2x + 1$	1	3	$(0; 1)$ and $(1; 3)$
$y = -x + 4$	4	3	$(0; 4)$ and $(1; 3)$
$y = 3x$	0	3	$(0; 0)$ and $(1; 3)$

$B = 0$ MEANS THE LINE PASSES THROUGH THE ORIGIN

Then $y = kx$ is a **direct proportion**: doubling x doubles y . Every proportional relationship met in Grade 6 was a linear function with $b = 0$.

What k and b do

COEFFICIENT	MEANING	ON THE GRAPH
$k > 0$	increasing	rises left to right
$k < 0$	decreasing	falls left to right
$k = 0$	constant	a horizontal line
b	the value at $x = 0$	where the line cuts the y -axis

The gradient is the change in y for a step of 1 in x :

$$k = \frac{y_2 - y_1}{x_2 - x_1}$$

TWO POINTS	WORKING	K
$(0; 1) \text{ and } (1; 3)$	$\frac{3 - 1}{1 - 0}$	2
$(0; 4) \text{ and } (1; 3)$	$\frac{3 - 4}{1 - 0}$	-1
$(1; 2) \text{ and } (4; 8)$	$\frac{8 - 2}{4 - 1}$	2

THE TWO DIFFERENCES MUST BE TAKEN IN THE SAME ORDER

$\frac{y_2 - y_1}{x_1 - x_2}$ gives the gradient with the wrong sign. Fix an order — second minus first, top and bottom — and keep it.

Meeting the axes

AXIS	SET	GIVES
y	$x = 0$	$y = b$
x	$y = 0$	$x = -\frac{b}{k}$

FUNCTION	CUTS OY AT	CUTS OX AT
$y = 2x + 1$	$(0; 1)$	$(-0.5; 0)$
$y = -x + 4$	$(0; 4)$	$(4; 0)$
$y = 3x - 6$	$(0; -6)$	$(2; 0)$

THE TWO INTERCEPTS ARE THE TWO EASIEST POINTS

For a line with $b \neq 0$ they are usually the quickest pair to plot, and they lie far apart, which makes the drawn line more accurate than one built from two nearby points.

Is the point on the line?

Substitute the coordinates. If the equation becomes true, the point lies on the line; if not, it does not.

LINE	POINT	CHECK	ON IT?
$y = 2x + 1$	$(3; 7)$	$2 \cdot 3 + 1 = 7$	yes
$y = 2x + 1$	$(3; 6)$	$7 \neq 6$	no
$y = -x + 4$	$(1; 3)$	$-1 + 4 = 3$	yes
$y = 3x$	$(2; 5)$	$6 \neq 5$	no

THIS IS WHAT THE NEXT BLOCK IS BUILT ON

A pair $(x; y)$ satisfying two linear equations at once lies on both lines — that is, at their crossing point. Systems of equations, three lessons from now, are exactly that idea.

02

Worked examples

EXAMPLE 1 Draw the graph of $y = 2x + 1$.

1

$x = 0$ gives $y = 1$.
The intercept.

2

$x = 1$ gives $y = 3$.

3

Plot $(0; 1)$ and $(1; 3)$ and join them.

4

Check with $x = -1$: $y = -1$ lies on the same line ✓

ANSWER

A line through $(0; 1)$ and $(1; 3)$

EXAMPLE 2 Find where $y = 3x - 6$ meets the axes.

$$① \quad x = 0: y = -6, \text{ so } (0; -6).$$

$$② \quad y = 0: 3x - 6 = 0.$$

$$③ \quad x = 2, \text{ so } (2; 0).$$

ANSWER $(0; -6)$ and $(2; 0)$ **EXAMPLE 3** A line passes through $(1; 2)$ and $(4; 8)$. Find its gradient.

$$① \quad k = \frac{y_2 - y_1}{x_2 - x_1}$$

$$② \quad = \frac{8 - 2}{4 - 1}$$

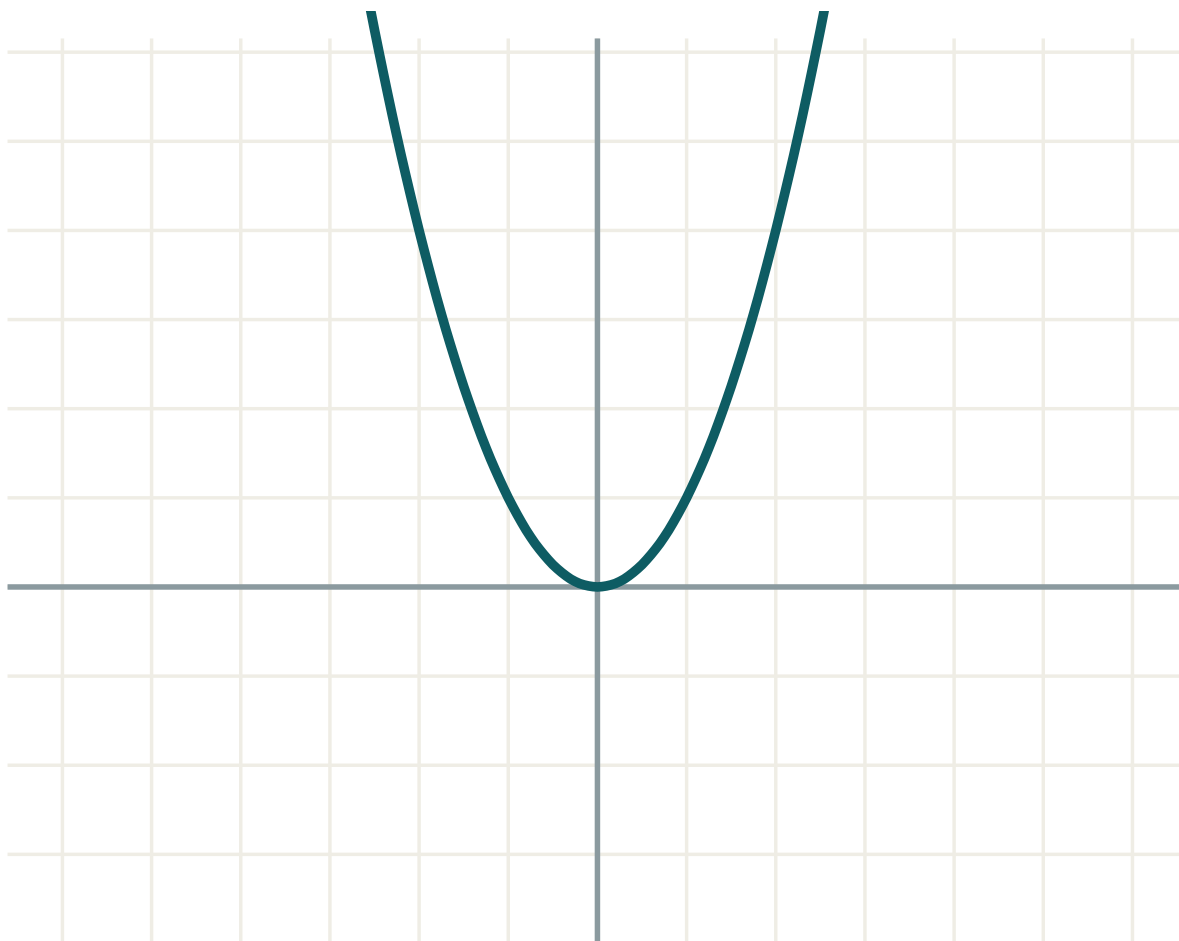
Same order top and bottom.

$$③ \quad = \frac{6}{3} = 2$$

ANSWER $k = 2$ **03 Interactive model**

Draw $y = 2x + 1$, $y = 2x$ and $y = 2x - 3$ on one grid; the class sees at once that k tilts the line and b slides it, without being told.

Move k and b and watch the line



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Linear function	Chiziqli funksiya	Линейная функция
Gradient	Burchak koeffitsiyenti	Угловой коэффициент
Intercept	Kesma	Свободный член
Straight line	To'g'ri chiziq	Прямая
Increasing	O'suvchi	Возрастающая
Decreasing	Kamayuvchi	Убывающая
Direct proportion	To'g'ri proporsionallik	Прямая пропорциональность
To lie on a line	Chiziqqa tegishli	Принадлежать прямой

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The graph of $y = kx + b$ is:

a curve

a straight line

a parabola

two lines

How many points determine it?

1

2

3

many

b is the value of y when:

$x = 1$

$x = 0$

$y = 0$

$k = 0$

$k < 0$ means the line:

rises

falls

is horizontal

is vertical

$y = 3x - 6$ meets Ox at:

$(0; -6)$

$(2; 0)$

$(-2; 0)$

$(6; 0)$

Is $(3; 6)$ on $y = 2x + 1$?

yes

no

only if k changes

cannot tell

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $y = 2x + 1$ at $x = 0$

ANSWER $y = 1$

2. $y = 2x + 1$ at $x = 1$

ANSWER $y = 3$

3. $y = -x + 4$ at $x = 1$

ANSWER $y = 3$

4. The intercept of $y = 3x - 6$

ANSWER $(0; -6)$

5. Is $y = 3x$ increasing?

ANSWER Yes

6. Is $y = -2x + 1$ increasing?

ANSWER No

7. The graph of $y = 5$ is

ANSWER a horizontal line

Medium · the standard question

7 PROBLEMS

1. $y = 3x - 6$ meets Ox at

ANSWER $(2; 0)$

2. $y = -x + 4$ meets Ox at

ANSWER $(4; 0)$

3. $y = 2x + 1$ meets Ox at

ANSWER $(-0.5; 0)$

4. The gradient through $(1; 2)$ and $(4; 8)$

ANSWER $k = 2$

5. The gradient through $(0; 4)$ and $(1; 3)$

ANSWER $k = -1$

6. Is $(3; 7)$ on $y = 2x + 1$?

ANSWER Yes

7. Is $(2; 5)$ on $y = 3x$?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. A line through $(0; 3)$ with gradient -2

ANSWER $y = -2x + 3$

2. A line through $(1; 5)$ and $(3; 11)$

ANSWER $y = 3x + 2$

3. For which k does $y = kx + 1$ pass through $(2; 7)$?

ANSWER $k = 3$

4. Where do $y = 2x + 1$ and $y = -x + 4$ cross?

ANSWER $(1; 3)$

5. Two lines with the same k are

ANSWER parallel

6. The graph of $y = kx$ always passes through

ANSWER the origin

7. For which b does $y = 2x + b$ pass through $(3; 1)$?

ANSWER $b = -5$

07 Homework

Homework — 5 tasks

Plot three points, not two — the third is your check.

1. Draw the graphs of $y = 2x - 3$ and $y = -x + 2$ on one grid.
2. Find where each of them meets the axes.
3. Find the gradient of the line through $(2; 1)$ and $(5; 10)$.
4. Decide whether $(4; 5)$ lies on $y = 2x - 3$.
5. Find b so that $y = 3x + b$ passes through $(2; 4)$.

Control work 10, and work on the mistakes

Coordinates, functions and the straight line in one paper.

LESSON 143–144

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 10

S8 9–11 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Plot and read points accurately and name their quadrants.
- Evaluate a function and solve for the argument.
- Draw a linear graph and find its intercepts.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 407–431
Cambridge	Stage 8, pp. 88–116
Workbook	Control paper M10

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Plot $A(3; 2)$, $B(-4; 1)$, $C(-2; -3)$, $D(5; -1)$ and name each quadrant	4	L137– 138
2	Find the length and midpoint of the segment from $(1; 3)$ to $(7; 3)$	4	L137– 138
3	For $f(x) = 2x + 1$ find $f(3)$, $f(-4)$ and the x with $f(x) = 11$	4	L139– 140
4	Give the domain of $y = \frac{1}{x - 3}$	3	L139– 140
5	Draw $y = 2x - 4$ and mark both intercepts	5	L141– 142
6	Find the gradient through $(1; 2)$ and $(4; 8)$, and say whether $(3; 6)$ lies on that line	5	L141– 142

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for labelling the points and one for the quadrant names; Q3 two for the substitutions and two for solving the equation; Q5 two for the two intercepts and one for a ruled line; Q6 two for the gradient and two for the substitution check.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
coordinates reversed	$(3; 2)$	along first, then up
quadrants numbered clockwise	$(-4; 1)$	anticlockwise from the top right
midpoint subtracted	$7 - 1 = 6$	$\frac{1 + 7}{2} = 4$
negative substituted carelessly	$f(-4) = 9$	$2 \cdot (-4) + 1 = -7$
domain left blank	“every number”	$x \neq 3$
line drawn freehand	a wobbling curve	a ruled line through the intercepts
gradient differences out of order	$\frac{8 - 2}{1 - 4} = -2$	$\frac{8 - 2}{4 - 1} = 2$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

BLOCK	THE SENTENCE
the plane	two axes, one origin, and an ordered pair for every point
quadrants	numbered anticlockwise, with a sign pattern each
functions	one input, exactly one output
three descriptions	formula, table and graph say the same thing
the domain	the values of x the rule allows
the linear function	$y = kx + b$
the graph	a straight line, two points enough, third as a check

LOOKING FORWARD

Two lines that cross give a pair of numbers satisfying both equations at once. That is a **system**, and it is the next block of the quarter.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $f(x) = 2x + 1$.

① $f(3) = 7$

② $f(-4) = 2 \cdot (-4) + 1 = -7$

The bracket keeps the sign right.

③ $2x + 1 = 11$

④ $x = 5$

ANSWER

7, -7, $x = 5$

EXAMPLE 2 Model answer, Q5: draw $y = 2x - 4$.

① $x = 0$: $y = -4$, so $(0; -4)$.

② $y = 0$: $2x = 4$, so $(2; 0)$.

③ Rule a line through both.

④ Check with $x = 3$: $y = 2$ lies on it ✓

ANSWER

A line through $(0; -4)$ and $(2; 0)$

EXAMPLE 3 Model answer, Q6: gradient through $(1; 2)$ and $(4; 8)$, and is $(3; 6)$ on it?

$$① \quad k = \frac{8 - 2}{4 - 1} = 2$$

$$② \quad \text{Through } (1; 2): 2 = 2 \cdot 1 + b, \text{ so } b = 0.$$

$$③ \quad \text{The line is } y = 2x.$$

$$④ \quad 2 \cdot 3 = 6 \checkmark \text{ — the point lies on it.}$$

ANSWER $k = 2$; yes**03** Interactive model

Return Q5 first and ask who used a ruler; the mark for a straight line is the easiest in the paper and the most often lost.

The chapter in twelve questions

The first coordinate is read:

up the y -axis

along the x -axis

from the origin

either way

$(-2; -3)$ lies in quadrant:

I

II

III

IV

$(0; 4)$ lies:

in quadrant I

on the y -axis

on the x -axis

at the origin

The midpoint of $(1; 3)$ and $(7; 3)$ is:

$(4; 3)$

$(3; 3)$

$(6; 3)$

$(8; 3)$

A function gives each input:

one output

two outputs

many outputs

none

For $f(x) = 2x + 1$, $f(-4)$ is:

9

-7

-9

7

The domain of $y = \underline{1}x - 3$ is:

every number

$x \neq 0$

$x \neq 3$

$x > 3$

The graph of $y = kx + b$ is:

a curve

a straight line

two lines

a circle

$y = 2x - 4$ cuts Oy at:

$(0; -4)$

$(2; 0)$

$(0; 2)$

$(-4; 0)$

And cuts Ox at:

$(0; -4)$

$(2; 0)$

$(4; 0)$

$(-2; 0)$

The gradient through $(1; 2)$ and $(4; 8)$ is:

2

-2

3

6

Is $(3; 6)$ on $y = 2x$?

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Coordinate	Koordinata	Координата
Quadrant	Chorak	Четверть
Function	Funksiya	Функция
Domain	Aniqlanish sohasi	Область определения
Gradient	Burchak koeffitsiyenti	Угловой коэффициент
Intercept	Kesma	Свободный член
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 also asks for:

the lengths

the quadrants

the gradients

the domain

Q2 midpoint uses:

the difference

the average

the product

the sum

Q3 last part is:

a substitution

an equation

a graph

a domain

Q4 answer is:

every number

$x \neq 3$

$x \neq 0$

$x > 0$

Q5 is drawn with:

freehand

a ruler

compasses

a protractor

Q6 checks the point by:

plotting only

substituting

measuring

guessing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The quadrant of $(3; 2)$

ANSWER I

2. The quadrant of $(-2; -3)$

ANSWER III

3. The midpoint of $(1; 3)$ and $(7; 3)$

ANSWER $(4; 3)$

4. $f(x) = 2x + 1$: $f(3)$

ANSWER 7

5. $f(-4)$

ANSWER -7

6. The domain of $y = \frac{1}{x - 3}$

ANSWER $x \neq 3$

7. $y = 2x - 4$ cuts Oy at

ANSWER $(0; -4)$

Medium · the standard question

7 PROBLEMS

1. $f(x) = 2x + 1 = 11$

ANSWER $x = 5$

2. $y = 2x - 4$ cuts Ox at

ANSWER $(2; 0)$

3. The gradient through $(1; 2)$ and $(4; 8)$

ANSWER 2

4. Is $(3; 6)$ on $y = 2x$?

ANSWER Yes

5. The length from $(1; 3)$ to $(7; 3)$

ANSWER 6

6. The equation of the line through $(1; 2)$ with gradient 2

ANSWER $y = 2x$

7. Where is $(6; 0)$?

ANSWER On the x -axis

Hard · stretch and reason

7 PROBLEMS

1. Where do $y = 2x - 4$ and $y = -x + 5$ cross?

ANSWER $(3; 2)$

2. A line through $(0; -4)$ and $(2; 0)$: its equation

ANSWER $y = 2x - 4$

3. For which k does $y = kx - 4$ pass through $(1; -1)$?

ANSWER $k = 3$

4. The domain of $y = \frac{1}{x^2 - 9}$

ANSWER $x \neq \pm 3$

5. The fourth vertex of a rectangle on $(1; 1)$, $(5; 1)$, $(5; 4)$

ANSWER $(1; 4)$

6. $f(x) = 2x + 1$: for which x is $f(x) = x$?

ANSWER $x = -1$

7. Two lines with equal gradients are

ANSWER parallel

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Plot $(4; 3)$, $(-1; 5)$, $(-3; -2)$ and $(2; -4)$ and name the quadrants.
3. For $f(x) = 3x - 2$ find $f(4)$, $f(-2)$ and the x with $f(x) = 10$.
4. Draw $y = -2x + 6$ and mark both intercepts.

5. Find the gradient of the line through $(-1; 1)$ and $(3; 9)$.

Systems of linear equations

Two equations, two unknowns, and a solution that is a pair — the point where two lines cross.

LESSON 145–146

2 × 40 MIN

MATEMATIKA 7, §62

S8 2.5 EXTENSION

LESSON CLOCK

14'

19'

23'

19'

5'

One equation, two unknowns	14 min
What a system is	19 min
Solving by drawing	23 min
One, none, or infinitely many	19 min
Homework	5 min

LEARNING OBJECTIVES

- Recognise a linear equation in two unknowns and describe its solutions.
- Understand a system, and that its solution is an ordered pair.
- Solve a system graphically.
- Say when a system has one solution, none, or infinitely many.

TEXTBOOK

National	pp. 432–439
Cambridge	Stage 8, pp. 31–36
Workbook	Extension sheet 2D

01

Explanation

One equation, two unknowns

$x + y = 5$ has no single answer: $(1; 4)$, $(2; 3)$, $(0; 5)$ and infinitely many other pairs all satisfy it.

x	0	1	2	3	5
$y = 5 - x$	5	4	3	2	0

Plotted, those pairs lie on a straight line. A linear equation in two unknowns **is** a line.

WHY ONE EQUATION IS NOT ENOUGH

One condition on two unknowns leaves one degree of freedom, and the answers fill a line. To pin the pair down, a second independent condition is needed — which is the whole idea of a system.

What a system is

$\{ x + y = 5 ; x - y = 1 \}$

A **system** asks for the pairs that satisfy **both** equations at once. Its solution is written as an ordered pair.

PAIR	$X + Y = 5$	$X - Y = 1$	A SOLUTION?
$(1; 4)$	✓	$-3 \neq 1$	no
$(3; 2)$	✓	✓	yes
$(4; 3)$	$7 \neq 5$	✓	no

THE ANSWER IS A PAIR, NOT A NUMBER

Writing “ $x = 3$ ” alone answers half the question. The solution is $(3; 2)$, and both numbers are needed for the mark.

Solving by drawing

Draw both lines on one grid; the point where they cross is the solution.

EQUATION	AS A FUNCTION	TWO POINTS
$x + y = 5$	$y = 5 - x$	$(0; 5), (5; 0)$
$x - y = 1$	$y = x - 1$	$(0; -1), (1; 0)$

The lines cross at $(3; 2)$, and substituting confirms it: $3 + 2 = 5$ ✓ and $3 - 2 = 1$ ✓.

A DRAWING GIVES AN ANSWER, NOT A PROOF

If the crossing falls between grid lines — say at $(2.4; 1.7)$ — the graph cannot say so exactly. That is why the next lesson introduces two algebraic methods.

One, none, or infinitely many

THE TWO LINES	PICTURE	SOLUTIONS	EXAMPLE
different gradients	they cross once	exactly one	$y = 2x, y = x + 1$
same gradient, different intercept	parallel	none	$y = 2x, y = 2x + 3$
same gradient and intercept	one line drawn twice	infinitely many	$y = 2x, 2y = 4x$

THE SAME THREE CASES AS FOR ONE EQUATION

A single linear equation had one root, no roots, or every number as a root. A system of two has one pair, no pair, or every point of a line. The pattern repeats one dimension higher.

02 Worked examples

EXAMPLE 1 Is $(3; 2)$ a solution of $\{x + y = 5; x - y = 1\}$?

① $3 + 2 = 5$ ✓

② $3 - 2 = 1$ ✓

③ Both hold, so yes.

ANSWER

Yes

EXAMPLE 2 Solve $\{x + y = 5 ; x - y = 1\}$ graphically.

① $y = 5 - x$ through $(0; 5)$ and $(5; 0)$.

② $y = x - 1$ through $(0; -1)$ and $(1; 0)$.

③ They cross at $(3; 2)$.

④ Check both equations ✓

ANSWER

$(3; 2)$

EXAMPLE 3 How many solutions has $\{y = 2x ; y = 2x + 3\}$?

① Both gradients are 2.

② The intercepts differ.

③ The lines are parallel and never meet.

④ No solutions.

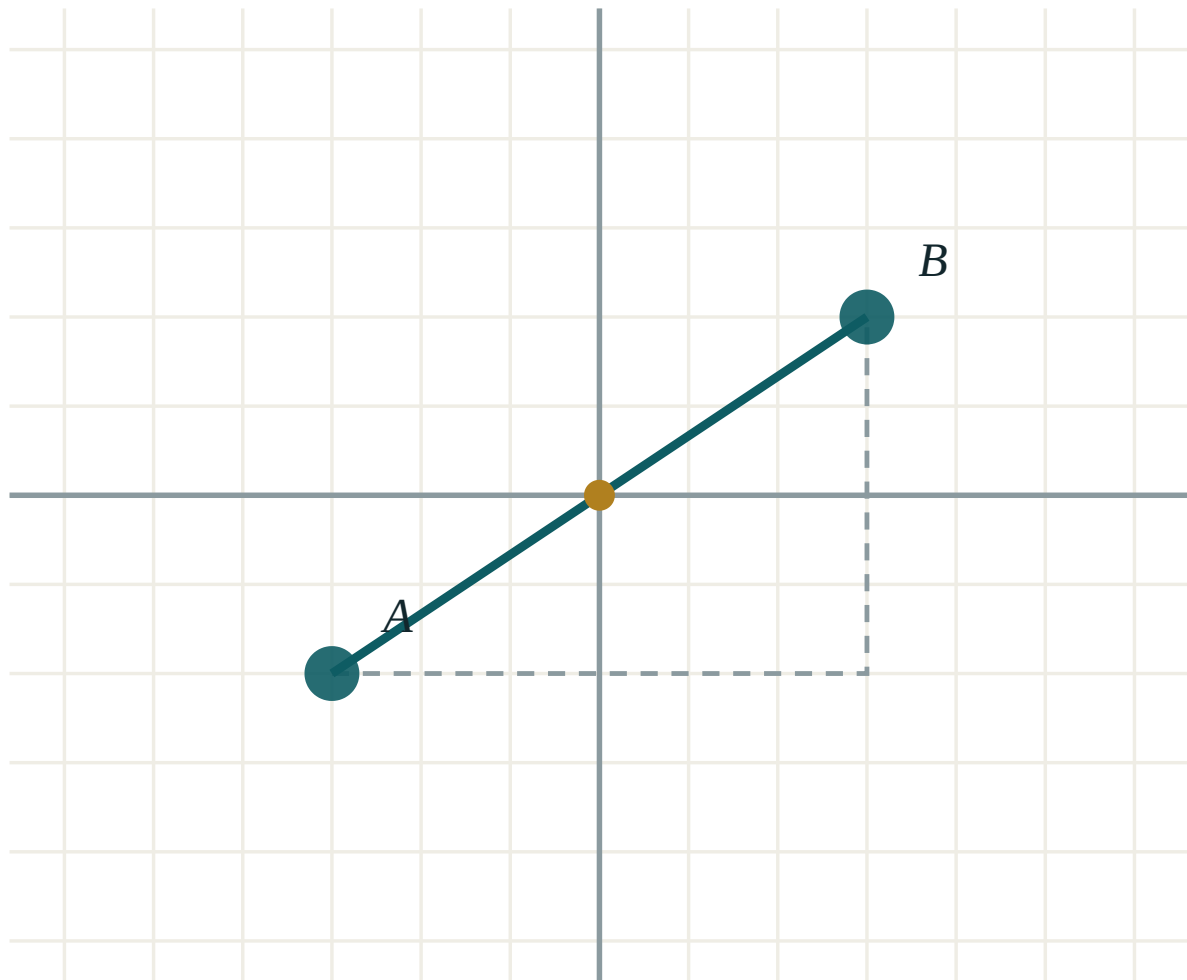
ANSWER

None

03 Interactive model

Ask for five pairs satisfying $x + y = 5$ before anything else; the class produces a line without knowing it, and the second equation then visibly picks one point out of it.

Two lines and their crossing



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
System of equations	Tenglamalar sistemasi	Система уравнений
Two unknowns	Ikki noma'lum	Два неизвестных
Solution of a system	Sistemaning yechimi	Решение системы
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
Graphical method	Grafik usul	Графический способ
Intersection point	Kesishish nuqtasi	Точка пересечения
Parallel	Parallel	Параллельные
Coincident	Ustma-ust tushuvchi	Совпадающие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$x + y = 5$ has:

one solution

no solutions

infinitely many pairs

two pairs

The graph of a linear equation in two unknowns is:

a point

a line

a curve

a region

The solution of a system is:

a number

an ordered pair

a line

an equation

Graphically it is:

the origin

the crossing point

the intercept

the gradient

Parallel lines mean:

one solution

no solutions

infinitely many

two solutions

Coincident lines mean:

one solution

no solutions

infinitely many

none of these

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Three pairs satisfying $x + y = 5$

ANSWER $(0; 5), (2; 3), (5; 0)$

2. Is $(3; 2)$ a solution of $x + y = 5$?

ANSWER Yes

3. Is $(3; 2)$ a solution of $x - y = 1$?

ANSWER Yes

4. Is $(1; 4)$ a solution of the system?

ANSWER No

5. $x + y = 5$ as a function

ANSWER $y = 5 - x$

6. $x - y = 1$ as a function

ANSWER $y = x - 1$

7. The solution of the system

ANSWER $(3; 2)$

Medium · the standard question

7 PROBLEMS

1. Solve $\{ y = 2x ; y = x + 1 \}$ graphically

ANSWER $(1; 2)$

2. How many solutions has $\{ y = 2x ; y = 2x + 3 \}$?

ANSWER None

3. How many has $\{ y = 2x ; 2y = 4x \}$?

ANSWER Infinitely many

4. Solve $\{ x + y = 6 ; x - y = 2 \}$

ANSWER $(4; 2)$

5. Solve $\{ y = x ; y = 4 - x \}$

ANSWER $(2; 2)$

6. Two lines with different gradients meet

ANSWER exactly once

7. Why is a graphical answer sometimes only approximate?

ANSWER The crossing may fall between grid lines

Hard · stretch and reason

7 PROBLEMS

1. Solve $\{ 2x + y = 7 ; x - y = 2 \}$

ANSWER $(3; 1)$

2. For which k has $\{ y = kx + 1 ; y = 2x + 3 \}$ no solution?

ANSWER $k = 2$

3. For which b has $\{ y = 2x + b ; y = 2x + 3 \}$ infinitely many?

ANSWER $b = 3$

4. Does $(2.5; 2.5)$ solve $\{ x + y = 5 ; x = y \}$?

ANSWER Yes

5. A system of two equations in two unknowns can have how many solutions?

ANSWER One, none, or infinitely many

6. Solve $\{ x + 2y = 8 ; x = 2 \}$

ANSWER $(2; 3)$

7. Write a system whose solution is $(1; 1)$

ANSWER e.g. $\{ x + y = 2 ; x - y = 0 \}$

07 Homework

Homework — 5 tasks

Always give the answer as a pair, and check it in both equations.

- Write five pairs satisfying $x + y = 7$ and plot them.
- Solve $\{ x + y = 7 ; x - y = 3 \}$ graphically.
- Decide how many solutions $\{ y = 3x + 1 ; y = 3x - 2 \}$ has, and why.
- Decide how many solutions $\{ y = 3x + 1 ; 2y = 6x + 2 \}$ has, and why.
- Check whether $(2; 5)$ solves $\{ 2x + y = 9 ; x - y = -3 \}$.

Methods of solving a system of linear equations

Substitution and elimination — two exact methods that finish where the drawing could only estimate.

LESSON 147–149

3 × 40 MIN

MATEMATIKA 7, §63

BEYOND STAGE 8

LESSON CLOCK

10

30'

35'

30'

15'

Two exact methods	10 min
Substitution	30 min
Elimination	35 min
Multiplying first	30 min
No solution, or infinitely many	15 min

LEARNING OBJECTIVES

- Solve a system by substitution.
- Solve a system by adding or subtracting the equations.
- Multiply an equation through to make elimination possible.
- Recognise from the algebra when there is no solution or infinitely many.

TEXTBOOK

National	pp. 440–450
Cambridge	Stage 8 extension
Workbook	Extension sheet 2E

01

Explanation

Two exact methods

The graph showed *where* the answer is; these two methods produce it exactly. Both work by turning two equations in two unknowns into one equation in one unknown.

METHOD	THE MOVE	BEST WHEN
substitution	express one unknown and put it into the other equation	one unknown already stands alone
elimination	add or subtract the equations to cancel an unknown	a pair of coefficients match or are opposite

BOTH METHODS HAVE THE SAME AIM

Get to one equation with one letter in it. Everything already known about linear equations then finishes the job, and the second unknown comes from back-substitution.

Substitution

STEP	WORKING FOR $\{ Y = X - 1 ; X + Y = 5 \}$
1 — express one unknown	$y = x - 1$
2 — substitute into the other	$x + (x - 1) = 5$
3 — solve	$2x = 6, x = 3$
4 — back-substitute	$y = 3 - 1 = 2$
5 — check both	$3 + 2 = 5 \checkmark, 3 - 2 = 1 \checkmark$

SUBSTITUTE INTO THE OTHER EQUATION

Putting $y = x - 1$ back into $y = x - 1$ gives $0 = 0$ and no information at all. The expression must go into the equation it did not come from.

Elimination

When the coefficients of one unknown are equal, subtract the equations; when they are opposite, add them.

SYSTEM	MOVE	RESULT	SOLUTION
$x + y = 5, x - y = 1$	add	$2x = 6$	$(3; 2)$
$2x + y = 7, x + y = 4$	subtract	$x = 3$	$(3; 1)$
$3x + 2y = 12, 3x - 2y = 0$	add	$6x = 12$	$(2; 3)$

SUBTRACTING MEANS SUBTRACTING EVERY TERM

From $2x + y = 7$ take $x + y = 4$: the right-hand sides give $7 - 4 = 3$, not 7. Line the two equations up and work column by column.

Multiplying first

When no coefficients match, multiply one equation — or both — through by a number until a pair of them does. Multiplying every term of an equation by the same non-zero number leaves its solutions untouched.

STEP	WORKING FOR $\{ 2X + 3Y = 12 ; X - Y = 1 \}$
multiply the second by 3	$3x - 3y = 3$
now the y coefficients are opposite	$+3y$
add the two equations	$5x = 15$
solve and back-substitute	$x = 3, y = 2$
check both	$6 + 6 = 12 \checkmark, 3 - 2 = 1 \checkmark$

Sometimes both equations need multiplying:

STEP	WORKING FOR $\{ 2X + 5Y = 16 ; 3X - 2Y = 5 \}$
multiply the first by 2	$4x + 10y = 32$
multiply the second by 5	$15x - 10y = 25$
add	$19x = 57$
solve and back-substitute	$x = 3, y = 2$

MULTIPLY EVERY TERM, BOTH SIDES INCLUDED

Turning $x - y = 1$ into $3x - 3y = 1$ changes the equation and loses the solution. The right-hand side is multiplied too.

No solution, or infinitely many

SYSTEM	THE ALGEBRA GIVES	MEANING
$y = 2x, y = 2x + 3$	$0 = 3$	no solution — parallel lines
$y = 2x, 2y = 4x$	$0 = 0$	infinitely many — the same line
$x + y = 5, x - y = 1$	$x = 3$	one solution

THE ALGEBRA REPORTS THE GEOMETRY

Reaching a false statement means the lines never meet; reaching an always-true one means they are the same line. Neither is a mistake — each is the answer, and it should be written in words.

02 Worked examples

EXAMPLE 1 Solve $\{y = x - 1 ; x + y = 5\}$ by substitution.

① y is already expressed.

② $x + (x - 1) = 5$
Into the other equation.

③ $2x = 6$, so $x = 3$.

④ $y = 2$. Check: $3 + 2 = 5$ ✓

ANSWER

$(3; 2)$

EXAMPLE 2 Solve $\{2x + y = 7 ; x + y = 4\}$ by elimination.

① The coefficients of y are equal.

② Subtract: $(2x - x) + (y - y) = 7 - 4$.

③ $x = 3$

④ $3 + y = 4$, so $y = 1$.

ANSWER

$(3; 1)$

EXAMPLE 3 Solve $\{ 2x + 3y = 12 ; x - y = 1 \}$.

- ① From the second, $x = y + 1$.
A coefficient of 1 invites substitution.

② $2(y + 1) + 3y = 12$

③ $5y = 10$, so $y = 2$.

④ $x = 3$. Check: $6 + 6 = 12$ ✓

ANSWER

$(3; 2)$

03 Interactive model

Solve one system by both methods side by side on the board; the class chooses its own favourite and, more usefully, learns to see which system suits which.

Which method, and what does it give?

$\{ y = x - 1 ; x + y = 5 \}$ suits:

substitution

elimination

drawing only

neither

$\{ x + y = 5 ; x - y = 1 \}$ suits:

substitution

elimination by adding

elimination by subtracting

drawing only

$\{ 2x + y = 7 ; x + y = 4 \}$ suits:

adding

subtracting

multiplying first

drawing

Adding $x + y = 5$ and $x - y = 1$ gives:

$$2x = 6$$

$$2y = 4$$

$$2x = 4$$

$$x = 6$$

Then x is:

2

3

4

6

And y is:

1

2

3

5

Reaching $0 = 3$ means:

one solution

no solution

infinitely many

an error

Reaching $0 = 0$ means:

one solution

no solution

infinitely many

an error

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Substitution method	O'rniga qo'yish usuli	Способ подстановки
Elimination method	Qo'shish usuli	Способ сложения
To express	Ifodalash	Выразить
To eliminate	Yo'qotmoq	Исключить
Opposite coefficients	Qarama-qarshi koeffitsiyentlar	Противоположные коэффициенты
To multiply through	Ko'paytirib chiqish	Умножить обе части
Back-substitution	Teskari o'rniga qo'yish	Обратная подстановка
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Substitution puts an expression into:

the same equation

the other equation

both

neither

Equal coefficients call for:

adding

subtracting

multiplying

drawing

Opposite coefficients call for:

adding

subtracting

dividing

nothing

$\{ 2x + y = 7 ; x + y = 4 \}$ has solution:

(3; 1)

(1; 3)

(2; 2)

(4; 0)

After finding one unknown you should:

stop

back-substitute

draw the graph

multiply

The final check uses:

one equation

both equations

the graph

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\{ y = x - 1 ; x + y = 5 \}$

ANSWER $(3; 2)$

2. $\{ x + y = 5 ; x - y = 1 \}$

ANSWER $(3; 2)$

3. $\{ 2x + y = 7 ; x + y = 4 \}$

ANSWER $(3; 1)$

4. $\{ y = 2x ; x + y = 6 \}$

ANSWER $(2; 4)$

5. $\{ x = 3 ; x + y = 8 \}$

ANSWER $(3; 5)$

6. $\{ y = 5 ; 2x + y = 11 \}$

ANSWER $(3; 5)$

7. Adding $x + y = 5$ and $x - y = 1$

ANSWER $2x = 6$

Medium · the standard question

7 PROBLEMS

1. $\{ 2x + 3y = 12 ; x - y = 1 \}$

ANSWER (3; 2)

2. $\{ 3x + 2y = 12 ; 3x - 2y = 0 \}$

ANSWER (2; 3)

3. $\{ x + 2y = 8 ; 3x - 2y = 0 \}$

ANSWER (2; 3)

4. $\{ 2x - y = 3 ; x + y = 3 \}$

ANSWER (2; 1)

5. $\{ y = 2x ; y = 2x + 3 \}$

ANSWER No solution

6. $\{ y = 2x ; 2y = 4x \}$

ANSWER Infinitely many

7. $\{ 4x + y = 14 ; x - y = 1 \}$

ANSWER (3; 2)

Hard · stretch and reason

7 PROBLEMS

1. $\{ 3x + 2y = 16 ; 5x - 2y = 8 \}$

ANSWER (3; 3.5)

2. $\{ 2x + 5y = 16 ; 3x - 2y = 5 \}$

ANSWER (3; 2)

3. $\{ 5x - 3y = 9 ; 2x + 3y = 12 \}$

ANSWER (3; 2)

4. For which k has $\{ y = kx + 1 ; y = 3x + 5 \}$ no solution?

ANSWER $k = 3$

5. $\{ \frac{x}{2} + y = 4 ; x - y = 2 \}$

ANSWER (4; 2)

6. Why does substituting an expression into its own equation fail?

ANSWER It gives $0 = 0$ and no information

7. $\{ x + y = 10 ; x - y = 10 \}$

ANSWER (10; 0)

07 Homework

Homework — 5 tasks

Say which method you chose and why, then check the pair in both equations.

1. Solve $\{ y = x + 2 ; 2x + y = 11 \}$ by substitution.
2. Solve $\{ x + y = 9 ; x - y = 3 \}$ by elimination.
3. Solve $\{ 3x + y = 11 ; x + y = 5 \}$.
4. Solve $\{ 2x + 3y = 13 ; x - y = -1 \}$.

5. Decide how many solutions $\{ y = 4x - 1 ; y = 4x + 2 \}$ has, and explain.

Solving problems with the help of systems

Two unknowns, two sentences, two equations — and an answer that names both quantities.

LESSON 150–152

3 × 40 MIN

MATEMATIKA 7, §64

BEYOND STAGE 8

LESSON CLOCK

10

30'

35'

30'

15'

When two unknowns are needed	10 min
Money and quantities	30 min
Numbers and digits	35 min
Geometry and motion	30 min
Checking and rejecting	15 min

LEARNING OBJECTIVES

- Choose two unknowns and say in words what each stands for.
- Turn two conditions into two equations.
- Solve the system by the method that suits it.
- Check both numbers against the original words and reject impossible answers.

TEXTBOOK

National	pp. 451–461
Cambridge	Stage 8 extension
Workbook	Extension sheet 2F

01

Explanation

When two unknowns are needed

If the problem names two quantities and gives two facts relating them, two letters and two equations are usually shorter than one letter and a long expression.

STEP	WHAT TO DO
1	let x and y be the two quantities, and say which is which
2	write the first condition as an equation
3	write the second condition as a second equation
4	solve the system
5	answer both parts of the question, with units

TWO FACTS, TWO EQUATIONS

Count the conditions in the problem before starting. Two unknowns need two independent facts; if only one is given, the problem has infinitely many answers and something has been misread.

Money and quantities

PROBLEM	UNKNOWN	SYSTEM	ANSWER
3 pens and 2 pencils cost 13 000; 1 pen and 1 pencil cost 5000	prices x and y	$3x + 2y = 13000, x + y = 5000$	pen 3000, pencil 2000
20 coins of 100 and 500 sum, worth 4400 in all	counts x and y	$x + y = 20, 100x + 500y = 4400$	14
a class of 30 with 4 more girls than boys	counts x and y	$x + y = 30, y - x = 4$	13

ONE EQUATION COUNTS, THE OTHER VALUES

In coin and ticket problems the first equation counts the items and the second adds up their worth. Recognising that pattern writes both equations almost automatically.

Numbers and digits

A two-digit number with tens digit a and units digit b is $10a + b$ — not ab , which would be a product.

STATEMENT	AS ALGEBRA
the number	$10a + b$
the sum of its digits	$a + b$
the number with its digits reversed	$10b + a$

Example: the digits of a two-digit number add to 9, and reversing them gives a number 27 larger.

STEP	WORKING
first condition	$a + b = 9$
second condition	$10b + a = 10a + b + 27$
simplify it	$9b - 9a = 27$, so $b - a = 3$
solve	$a = 3, b = 6$
the number	36

DIGITS ARE WHOLE AND THE TENS DIGIT IS NOT ZERO

A root of $a = 4.5$ or $a = 0$ must be rejected: no such two-digit number exists. The algebra does not know this — the reader does.

Geometry and motion

PROBLEM	SYSTEM	ANSWER
a rectangle of perimeter 36 whose length exceeds its width by 4	$2(x + y) = 36, y - x = 4$	7 by 11
two angles are complementary and differ by 20°	$x + y = 90, y - x = 20$	35° and 55°
a boat goes 20 km/h downstream and 12 km/h up	$x + y = 20, x - y = 12$	boat 16, current 4

DOWNSTREAM AND UPSTREAM

The current adds its speed going down and subtracts it coming up, so the two speeds are $x + y$ and $x - y$. Adding the equations gives the boat's own speed; subtracting gives the current's.

Checking and rejecting

ANSWER	QUANTITY	VERDICT
$(3000; 2000)$	two prices	accepted
$(14; 6)$	two counts of coins	accepted — both whole
$(4.5; 4.5)$	two counts of pupils	rejected — not whole
$(-2; 8)$	two lengths	rejected — negative

CHECK IN THE WORDS, NOT IN THE EQUATIONS

Substituting into your own equations only tests your arithmetic. Reading the answer back into the original sentences also tests the translation, which is where the marks are.

02

Worked examples

EXAMPLE 1 3 pens and 2 pencils cost 13 000 sum; a pen and a pencil together cost 5000. Find each price.

- 1 Let x be the price of a pen and y of a pencil.
- 2 $3x + 2y = 13\ 000$ and $x + y = 5000$.
- 3 From the second, $y = 5000 - x$; substitute: $3x + 10\ 000 - 2x = 13\ 000$.
- 4 $x = 3000$, so $y = 2000$. Check: $9000 + 4000 = 13\ 000$ ✓

ANSWER

Pen 3000, pencil 2000 sum

EXAMPLE 2 The digits of a two-digit number add to 9, and reversing them gives a number 27 larger. Find it.

- 1 Let the digits be a (tens) and b (units).
- 2 $a + b = 9$
- 3 $10b + a = 10a + b + 27$, which is $b - a = 3$.
- 4 $a = 3, b = 6$ — the number is 36, and $63 - 36 = 27$ ✓

ANSWER

36

EXAMPLE 3 A boat travels at 20 km/h downstream and 12 km/h upstream. Find the speed of the boat and of the current.

- 1 Let x be the boat's speed and y the current's.
- 2 $x + y = 20$ and $x - y = 12$.
- 3 Add: $2x = 32$, so $x = 16$.
- 4 $y = 4$. Check: $16 + 4 = 20$ ✓ and $16 - 4 = 12$ ✓

ANSWER

Boat 16 km/h, current 4 km/h

03 Interactive model

Ask the class to write only the two equations for six problems and collect them before any solving; almost every lost mark in this block is a translation error, not an algebraic one.

Write the two equations

A pen x and a pencil y cost 5000 together:

$$x + y = 5000$$

$$xy = 5000$$

$$x - y = 5000$$

$$x = 5000y$$

Three pens and two pencils cost 13 000:

$$3x + 2y = 13000$$

$$5xy = 13000$$

$$3x = 2y$$

$$x + y = 13000$$

20 coins altogether:

$$x + y = 20$$

$$xy = 20$$

$$100x + 500y = 20$$

$$x - y = 20$$

Their value is 4400 sum:

$$x + y = 4400$$

$$100x + 500y = 4400$$

$$600xy = 4400$$

$$x = 4400$$

A two-digit number with digits a and b is:

$$ab$$

$$a + b$$

$$10a + b$$

$$10b + a$$

Reversed, it is:

$$10a + b$$

$$10b + a$$

$$a + b$$

$$ba$$

A rectangle of perimeter 36:

$$x + y = 36$$

$$2(x + y) = 36$$

$$xy = 36$$

$$x - y = 36$$

Downstream speed with current y :

$$x - y$$

$$x + y$$

$$xy$$

$$x$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Two unknowns	Ikki noma'lum	Два неизвестных
Condition	Shart	Условие
To denote	Belgilamoq	Обозначить
Total cost	Umumiy narx	Общая стоимость
Digit	Raḡam	Цифра
Two-digit number	Ikki xonali son	Двузначное число
Perimeter	Perimetr	Периметр
To interpret	Talqin qilmoq	Истолковать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Two unknowns need how many independent facts?

A two-digit number with digits a , b is:

Pens x and pencils y : $3x + 2y = 13\ 000$ and $x + y = 5000$ give a pen at:

Complementary angles differing by 20° are:

30° and 50°

35° and 55°

40° and 60°

45° and 65°

Downstream 20, upstream 12: the current is:

4

8

16

32

An answer of 4.5 pupils is:

accepted

rejected

rounded up

doubled

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Two numbers add to 12 and differ by 4

ANSWER 8 and 4

2. A class of 30 with 4 more girls than boys

ANSWER 17 girls, 13 boys

3. Complementary angles differing by 20°

ANSWER 55° and 35°

4. A two-digit number with digits a, b

ANSWER $10a + b$

5. The same number reversed

ANSWER $10b + a$

6. Downstream speed with current y

ANSWER $x + y$

7. Upstream speed

ANSWER $x - y$

Medium · the standard question

7 PROBLEMS

1. 3 pens and 2 pencils cost 13 000; one of each costs 5000

ANSWER Pen 3000, pencil 2000

2. 20 coins of 100 and 500 worth 4400

ANSWER 14 of 100, 6 of 500

3. A rectangle of perimeter 36, length 4 more than width

ANSWER 7 by 11

4. Downstream 20, upstream 12

ANSWER Boat 16, current 4

5. Digits add to 9, reversal is 27 larger

ANSWER 36

6. Two numbers: three times the first plus the second is 11, and they add to 5

ANSWER (3; 2)

7. Supplementary angles differing by 40°

ANSWER 110° and 70°

Hard · stretch and reason

7 PROBLEMS

1. Digits add to 11 and the reversal is 27 smaller

ANSWER 74

2. 5 apples and 3 pears cost 19 000; 2 apples and 4 pears cost 16 000

ANSWER Apple 2000, pear 3000

3. A boat covers 60 km down in 3 h and 60 km up in 5 h

ANSWER Boat 16, current 4

4. Two angles of a triangle differ by 30° and the third is 60°

ANSWER 45° and 75°

5. A father is 4 times his son and their ages add to 50

ANSWER Son 10, father 40

6. Why must a digit answer be whole and under 10?

ANSWER Digits are the symbols 0 to 9

7. Two numbers have sum 20 and one is 4 times the other

ANSWER 4 and 16

07 Homework

Homework — 5 tasks

Write both “let” sentences in words, and answer with both quantities.

1. Two numbers add to 26 and differ by 8. Find them.
2. 4 notebooks and 3 pens cost 26 000 sum; 2 notebooks and 1 pen cost 12 000. Find each price.
3. The digits of a two-digit number add to 12 and the reversal is 18 larger. Find the number.
4. A rectangle has perimeter 48 cm and its length is 6 cm more than its width. Find its sides.
5. A boat does 22 km/h downstream and 14 km/h upstream. Find both speeds.

Control work 11, and work on the mistakes

Systems of two linear equations, by both methods, and the problems behind them.

LESSON 153–154

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 11

EXTENSION REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Solve a system by substitution and by elimination.
- Say from the algebra when there is no solution or infinitely many.
- Turn a two-condition problem into a system and answer both parts.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 432–461
Cambridge	Stage 8 extension
Workbook	Control paper M11

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Solve $\{ y = x - 1 ; x + y = 5 \}$ by substitution	4	L147– 149
2	Solve $\{ x + y = 9 ; x - y = 3 \}$ by elimination	4	L147– 149
3	Solve $\{ 2x + 3y = 12 ; x - y = 1 \}$	5	L147– 149
4	How many solutions has $\{ y = 4x - 1 ; y = 4x + 2 \}$? Explain	3	L145– 149
5	Two numbers add to 26 and differ by 8: find them	4	L150– 152
6	A boat does 22 <i>km/h</i> downstream and 14 <i>km/h</i> upstream: find both speeds	5	L150– 152

WHERE THE MARKS ACTUALLY GO

Q1 one mark for back-substituting rather than stopping at x ; Q2 two for lining the equations up correctly; Q4 two for the words “parallel lines, so no solution”; Q5 and Q6 one each for the “let” sentences and one each for answering both parts.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
substituted into its own equation	$0 = 0$	put it into the other one
only one unknown found	$x = 3$	$(3; 2)$
right-hand sides not subtracted	$2x = 7$	$2x = 7 - 4$
multiplied one side only	$3x - 3y = 1$	$3x - 3y = 3$
“no solution” called an error	the working abandoned	$0 = 3$
unknowns not declared	two equations with no “let”	say what x and y are
only one quantity reported	“the boat does 18”	both the boat and the current

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

IDEA	THE SENTENCE
one linear equation in two unknowns	its solutions fill a straight line
a system	the pairs satisfying both equations at once
graphically	the crossing point of the two lines
substitution	express one unknown, put it into the other equation
elimination	add or subtract to cancel an unknown
$0 = 3$	parallel lines — no solution
$0 = 0$	the same line — infinitely many

LOOKING FORWARD

The last block of the year is combinatorics: counting arrangements and selections. It needs no equations at all — only the two rules of “and” and “or”.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $\{ 2x + 3y = 12 ; x - y = 1 \}$.

- ① From the second, $x = y + 1$.
A coefficient of 1.

- ② $2(y + 1) + 3y = 12$

- ③ $5y = 10$, so $y = 2$.

- ④ $x = 3$. Check: $6 + 6 = 12$ ✓ and $3 - 2 = 1$ ✓

ANSWER

$(3; 2)$

EXAMPLE 2 Model answer, Q4: $\{ y = 4x - 1 ; y = 4x + 2 \}$.

- ① Setting the right sides equal: $4x - 1 = 4x + 2$.

- ② $0 = 3$

- ③ That is false for every x .

- ④ The gradients are equal and the intercepts differ: parallel lines, no solution.

ANSWER

No solution

EXAMPLE 3 Model answer, Q6: downstream 22, upstream 14.

- 1 Let x be the boat's own speed and y the current's.
- 2 $x + y = 22$ and $x - y = 14$.
- 3 Add: $2x = 36$, so $x = 18$.
- 4 $y = 4$. Both speeds are asked for, so both are given.

ANSWER

Boat 18 km/h, current 4 km/h

03 Interactive model

Return Q5 and Q6 first and count how many scripts reported only one number; the habit of answering the whole question is worth more marks than any technique.

The chapter in twelve questions

$x + y = 5$ alone has:

one solution

no solutions

a line of solutions

two solutions

The solution of a system is:

a number

an ordered pair

a line

an equation

Graphically it is:

the intercept

the crossing point

the origin

the gradient

$\{ y = x - 1 ; x + y = 5 \}$ gives:

$(3; 2)$

$(2; 3)$

$(4; 3)$

$(1; 4)$

Adding $x + y = 9$ and $x - y = 3$ gives:

$2x = 12$

$2y = 6$

$2x = 6$

$x = 12$

So the solution is:

$(6; 3)$

$(3; 6)$

$(12; 3)$

$(6; 6)$

$\{ 2x + 3y = 12 ; x - y = 1 \}$ gives:

$(2; 3)$

$(3; 2)$

$(1; 2)$

$(4; 3)$

Equal gradients and different intercepts give:

one solution

no solution

infinitely many

two solutions

Equal gradients and equal intercepts give:

one solution

no solution

infinitely many

none of these

Two numbers adding to 26 and differing by 8 are:

17 and 9

16 and 10

18 and 8

15 and 11

Downstream 22 and upstream 14 give a current of:

4

8

18

36

When a problem names two quantities you should report:

one of them

both of them

the equations

x only

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
System	Sistema	Система
Substitution	O'rniga qo'yish	Подстановка
Elimination	Qo'shish usuli	Способ сложения
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
No solution	Yechimga ega emas	Нет решений
Infinitely many	Cheksiz ko'p	Бесконечно много
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 finishes with:

Q2 uses:

Q3 is quickest by:

Q4 answer is:

one solution

no solution

infinitely many

$(0; 3)$

Q5 needs:

one number

two numbers

an equation

a graph

Q6 asks for:

the boat only

the current only

both speeds

the distance

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\{ y = x - 1 ; x + y = 5 \}$

ANSWER (3; 2)

2. $\{ x + y = 9 ; x - y = 3 \}$

ANSWER (6; 3)

3. $\{ 2x + 3y = 12 ; x - y = 1 \}$

ANSWER (3; 2)

4. $\{ y = 4x - 1 ; y = 4x + 2 \}$

ANSWER No solution

5. Two numbers adding to 26 and differing by 8

ANSWER 17 and 9

6. Downstream 22, upstream 14: the boat

ANSWER 18

7. And the current

ANSWER 4

Medium · the standard question

7 PROBLEMS

1. $\{ 3x + y = 11 ; x + y = 5 \}$

ANSWER (3; 2)

2. $\{ y = 2x + 1 ; y = 3x - 2 \}$

ANSWER (3; 7)

3. $\{ 2x + y = 8 ; 3x - y = 7 \}$

ANSWER (3; 2)

4. $\{ y = 4x - 1 ; 2y = 8x - 2 \}$

ANSWER Infinitely many

5. A rectangle of perimeter 48, length 6 more than width

ANSWER 9 by 15

6. Complementary angles differing by 20°

ANSWER 55° and 35°

7. $\{ x + 2y = 7 ; x = 3 \}$

ANSWER (3; 2)

Hard · stretch and reason

7 PROBLEMS

1. $\{ 2x + 5y = 16 ; 3x - 2y = 5 \}$

ANSWER (3; 2)

2. $\{ 4x + 3y = 18 ; 2x - y = 4 \}$

ANSWER (3; 2)

3. Digits add to 12, reversal 18 larger

ANSWER 57

4. 4 notebooks and 3 pens cost 26 000; 2 notebooks and 1 pen cost 12 000

ANSWER Notebook 5000, pen 2000

5. For which k has $\{ y = kx + 1 ; y = 5x - 2 \}$ no solution?

ANSWER $k = 5$

6. Why is " $x = 3$ " an incomplete answer to a system?

ANSWER The solution is a pair

7. $\{ x - y = 0 ; x + y = 14 \}$

ANSWER (7; 7)

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Solve $\{ y = 2x - 3 ; x + y = 6 \}$ by substitution.
3. Solve $\{ 3x + 2y = 16 ; 3x - 2y = 8 \}$ by elimination.
4. Say how many solutions $\{ y = 5x + 2 ; 2y = 10x + 4 \}$ has and why.
5. Two numbers add to 34 and differ by 10. Find them.

The basic rules of combinatorics

Two rules decide every counting problem: “and” multiplies, “or” adds.

LESSON 155–157

3 × 40 MIN

MATEMATIKA 7, §65

S8 13

LESSON CLOCK

15'

30'

30'

30'

15'

Counting without listing	15 min
The multiplication rule	30 min
The addition rule	30 min
Trees and tables	30 min
With or without repetition	15 min

LEARNING OBJECTIVES

- State and apply the multiplication rule and the addition rule.
- Decide from the wording which rule a problem needs.
- Draw a tree diagram and a table of outcomes.
- Count outcomes with and without repetition.

TEXTBOOK

National	pp. 462–472
Cambridge	Stage 8, pp. 126–134
Workbook	Exercise 13.1

01

Explanation

Counting without listing

How many two-course meals can be made from 3 soups and 4 main dishes? The list has 12 lines; the answer can be had without writing any of them.

PROBLEM	BY LISTING	BY RULE
3 soups, 4 mains	12 lines	$3 \cdot 4$
10 digits, 4-figure code	10 000 lines	10^4
26 letters, 3-letter word	17 576 lines	26^3

WHY THE RULES MATTER

Listing works for a dozen outcomes and fails for a thousand. Combinatorics is the art of counting a set without writing it down — which is also what makes probability possible.

The multiplication rule

If a first choice can be made in m ways and, after it, a second in n ways, then the pair of choices can be made in $m \cdot n$ ways.

“and” $\implies m \cdot n$

PROBLEM	WORKING	ANSWER
3 shirts and 4 ties	$3 \cdot 4$	12
5 starters, 3 mains, 2 puddings	$5 \cdot 3 \cdot 2$	30
a coin and a die	$2 \cdot 6$	12
two dice	$6 \cdot 6$	36

THE RULE EXTENDS TO ANY NUMBER OF STAGES

Three choices give $m \cdot n \cdot p$, four give $m \cdot n \cdot p \cdot q$. Each new stage multiplies, which is why counting problems grow so quickly.

The addition rule

If one thing can be chosen in m ways and another, incompatible with it, in n ways, then one **or** the other can be chosen in $m + n$ ways.

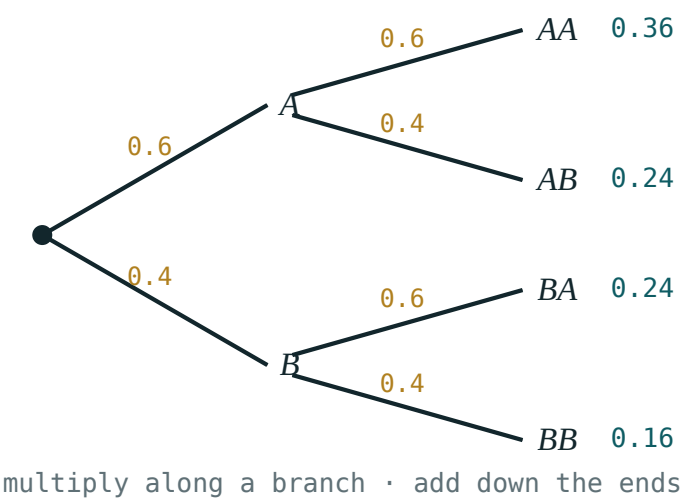
“or” $\implies m + n$

PROBLEM	WORKING	ANSWER
one book from 7 novels or 5 poetry books	$7 + 5$	12
travel by one of 3 buses or 2 trains	$3 + 2$	5
a shirt and a tie, or a jumper alone (3, 4, 6)	$3 \cdot 4 + 6$	18

THE ADDITION RULE NEEDS THE CHOICES TO EXCLUDE EACH OTHER

If 2 of the 7 novels are also counted among the poetry books, adding gives 12 where the true count is 10. Check that nothing is counted twice before adding.

Trees and tables



A tree for two stages of choice

A tree diagram shows every outcome as a path; the number of paths is the product of the branchings — the multiplication rule made visible.

TOOL	GOOD FOR		READS AS			
tree diagram	two or three stages		multiply along a path			
table of outcomes	exactly two stages		rows times columns			
the rules alone	many stages, or large numbers		a single product			

COIN \ DIE	1	2	3	4	5	6
heads	H1	H2	H3	H4	H5	H6
tails	T1	T2	T3	T4	T5	T6

With or without repetition

The second question to ask, after “and or or?”, is whether an item may be used twice.

PROBLEM	REPETITION	WORKING	ANSWER
3-figure codes from 10 digits	allowed	$10 \cdot 10 \cdot 10$	1000
3-figure codes, digits all different	not allowed	$10 \cdot 9 \cdot 8$	720
seating 4 pupils in 4 chairs	not allowed	$4 \cdot 3 \cdot 2 \cdot 1$	24
2-letter codes from 5 letters	allowed	$5 \cdot 5$	25

WITHOUT REPETITION THE NUMBERS COUNT DOWN

10, 9, 8 — each stage has one fewer choice because one item has been used. That descending product is the whole difference between the two cases.

02 Worked examples

EXAMPLE 1 A canteen offers 5 starters, 3 main dishes and 2 puddings. How many three-course meals are possible?

① The three choices are made together — “and”.

② $5 \cdot 3 \cdot 2$

③ $= 30$

ANSWER

30

EXAMPLE 2 How many 3-figure codes can be made from 10 digits if the digits must all differ?

① First digit: 10 ways.

② Second: 9, since one is used.

③ Third: 8.

④ $10 \cdot 9 \cdot 8 = 720$

ANSWER

720

EXAMPLE**3**

A pupil may take one of 7 novels or one of 5 poetry books, with no book in both lists. How many choices?

① The two options exclude each other — “or”.

② $7 + 5$

③ $= 12$

ANSWER**12****03 Interactive model**

Have the class list all 12 shirt-and-tie outfits by hand, then ask for the number with 8 shirts and 9 ties; the rule is welcomed rather than imposed.

And, or, and repeats

3 shirts **and** 4 ties give:

7 novels **or** 5 poetry books give:

A coin and a die give:

Two dice give:

12

24

36

6

3-figure codes from 10 digits, repeats allowed:

30

720

1000

120

The same with all digits different:

720

1000

504

90

Seating 4 pupils in 4 chairs:

4

12

16

24

The addition rule requires the options to be:

equal

mutually exclusive

ordered

repeated

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Combinatorics	Kombinatorika	Комбинаторика
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Addition rule	Qo'shish qoidasi	Правило сложения
Outcome	Natija	Исход
Tree diagram	Daraxt sxemasi	Дерево вариантов
With repetition	Takrorlanish bilan	С повторением
Without repetition	Takrorlanishsiz	Без повторения
Mutually exclusive	Birgalikda bo'lmagan	Взаимоисключающие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“And” calls for:

adding

multiplying

subtracting

dividing

“Or” calls for:

adding

multiplying

subtracting

dividing

5 starters, 3 mains, 2 puddings give:

10

30

60

15

Without repetition the numbers:

stay the same

count down

count up

double

3-figure codes with different digits number:

720

1000

30

120

A tree diagram shows each outcome as:

a branch

a path

a node

a row

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 3 shirts and 4 ties

ANSWER 12

2. 7 novels or 5 poetry books

ANSWER 12

3. A coin and a die

ANSWER 12

4. Two dice

ANSWER 36

5. Two coins

ANSWER 4

6. 5 starters, 3 mains, 2 puddings

ANSWER 30

7. 3 buses or 2 trains

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. 3-figure codes from 10 digits, repeats allowed

ANSWER 1000

2. The same, all digits different

ANSWER 720

3. Seating 4 pupils in 4 chairs

ANSWER 24

4. 2-letter codes from 5 letters, repeats allowed

ANSWER 25

5. The same without repeats

ANSWER 20

6. A shirt and a tie (3, 4) or a jumper (6)

ANSWER 18

7. Three coins

ANSWER 8

Hard · stretch and reason

7 PROBLEMS

1. Car plates: 3 letters from 26 then 3 digits, repeats allowed

ANSWER $26^3 \cdot 10^3 = 17\,576\,000$

2. How many 4-figure PINs have all digits different?

ANSWER 5040

3. How many 3-digit numbers are there in all?

ANSWER 900

4. How many 3-digit numbers have no repeated digit?

ANSWER 648

5. How many even 3-figure codes from 10 digits, repeats allowed?

ANSWER 500

6. Why does the addition rule fail if the lists overlap?

ANSWER The common items are counted twice

7. Arrangements of the 5 letters of a word with all letters different

ANSWER 120

07 Homework

Homework — 5 tasks

Ask two questions first: “and or or?”, then “may anything repeat?”

1. A shop has 6 shirts and 5 ties. How many outfits?
2. How many 4-figure codes can be made from 10 digits with repeats allowed, and how many without?
3. A pupil chooses one of 8 novels or one of 6 science books. How many choices?
4. Draw a tree diagram for tossing three coins and count the outcomes.

5. In how many ways can 5 pupils be seated in 5 chairs?

Kinds of combinatorial problem

Arrangements, permutations and selections — and the one question that tells them apart.

LESSON 158–160

3 × 40 MIN

MATEMATIKA 7, §66

S8 13

LESSON CLOCK

15'

30'

30'

30'

15'

The question that decides everything	15 min
Permutations and the factorial	30 min
Arrangements of k from n	30 min
Selections	30 min
Telling them apart	15 min

LEARNING OBJECTIVES

- Ask whether order matters, and answer accordingly.
- Count permutations of n objects with $n!$.
- Count arrangements of k objects chosen from n .
- Count selections where order does not matter.

TEXTBOOK

National	pp. 473–483
Cambridge	Stage 8, pp. 126–134
Workbook	Exercise 13.2

01

Explanation

The question that decides everything

Before counting anything, ask: **does the order matter?** Two problems with identical numbers can have different answers because of it.

PROBLEM	ORDER?	BECAUSE
first and second prize among 5 pupils	yes	first is not second
two pupils on duty from 5	no	the pair is the same either way
a 4-figure PIN	yes	$1234 \neq 4321$
a hand of 4 cards	no	the hand is a set

ORDER DOUBLES OR HALVES THE ANSWER

From 5 pupils, ordered pairs number 20 and unordered pairs 10. Every ordered pair appears twice in the ordered count — once each way round.

Permutations and the factorial

A **permutation** arranges all n objects in a row. There are n choices for the first place, $n - 1$ for the second, and so on.

$$n! = n \cdot (n - 1) \cdot \dots \cdot 2 \cdot 1$$

N	$N!$	MEANING
1	1	one object, one order
2	2	AB, BA
3	6	six orders of three letters
4	24	seating four pupils
5	120	five books on a shelf
6	720	growing fast

THE FACTORIAL GROWS FASTER THAN ANYTHING MET SO FAR

$10! = 3\,628\,800$. Ten books can be shelved in more than three and a half million orders — a good reason never to try listing them.

Arrangements of k from n

When only k of the n objects are used and the order matters, the product simply stops after k factors.

$$n \cdot (n - 1) \cdot \dots \text{--- } k \text{ factors}$$

PROBLEM	WORKING	ANSWER
first and second prize among 5	$5 \cdot 4$	20
gold, silver and bronze among 8	$8 \cdot 7 \cdot 6$	336
3 of 10 digits, all different	$10 \cdot 9 \cdot 8$	720
president and secretary from 12	$12 \cdot 11$	132

A PERMUTATION IS THE CASE $k = n$

Arranging 5 of 5 gives $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5!$. The two ideas are one idea, stopped at different points.

Selections

When the order does **not** matter, count the ordered arrangements first and then divide by the number of orders of the chosen group.

selections = $\frac{\text{arrangements}}{k!}$

PROBLEM	ORDERED	DIVIDE BY	ANSWER
2 pupils on duty from 5	$5 \cdot 4 = 20$	$2! = 2$	10
3 pupils from 6	$6 \cdot 5 \cdot 4 = 120$	$3! = 6$	20
handshakes among 10 people	$10 \cdot 9 = 90$	2	45
diagonals and sides of a hexagon	$6 \cdot 5 = 30$	2	15

HANDSHAKES, DIAGONALS AND PAIRS ARE ONE PROBLEM

Every pair of people shakes once, every pair of vertices joins once. The hexagon's 15 pairs are its 6 sides and 9 diagonals — a check worth making.

Telling them apart

WORDING	ORDER?	KIND
“in how many orders”	yes	permutation
“first, second and third”	yes	arrangement
“a code” or “a number”	yes	arrangement
“choose a group of”	no	selection
“how many pairs”	no	selection
“how many handshakes”	no	selection

PRIZES ARE ORDERED; DUTIES USUALLY ARE NOT

“First and second prize” distinguishes the two places; “two pupils on duty” does not. Reading the wording carefully is the whole of the skill — the arithmetic afterwards is easy.

02

Worked examples

EXAMPLE 1 In how many ways can 5 books be arranged on a shelf?

1

All five are used and the order matters.

2

$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$

3

$= 120$

ANSWER

120

EXAMPLE 2 From 8 runners, in how many ways can gold, silver and bronze be awarded?

① Three of eight, and the order matters.

② $8 \cdot 7 \cdot 6$

③ $= 336$

ANSWER

336

EXAMPLE 3 How many handshakes take place if 10 people each shake hands once with every other?

① Ordered pairs: $10 \cdot 9 = 90$.

② Each handshake was counted twice.

③ $90 \div 2 = 45$

ANSWER

45

03 Interactive model

Have five pupils stand at the front and ask first for prize orders, then for duty pairs; the same five children give 20 and 10, and the distinction is made physically.

Does the order matter?

First and second prize among 5:

order matters

order does not

neither

both

So the count is:

10

20

25

120

Two pupils on duty from 5:

order matters

order does not

neither

both

So the count is:

10

20

25

120

5! equals:

25

60

120

720

Gold, silver and bronze among 8:

24

56

336

512

Handshakes among 10 people:

45

90

100

20

Choosing 3 pupils from 6:

18

20

120

720

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Arrangement	O'rinlashtirish	Размещение
Permutation	O'rin almashtirish	Перестановка
Selection	Tanlash	Сочетание
Factorial	Faktorial	Факториал
Order matters	Tartib muhim	Порядок важен
Order does not matter	Tartib muhim emas	Порядок не важен
Committee	Guruh	Комиссия
Handshake	Qo'l berish	Рукопожатие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A permutation arranges:

some of the objects

all of the objects

two objects

none

$4!$ equals:

12

16

24

64

Order matters in:

a committee

a PIN

a handshake

a pair

Order does not matter in:

a race result

a code

a group on duty

a queue

To go from arrangements to selections you:

multiply by $k!$

divide by $k!$

add k

do nothing

Handshakes among n people number:

n^2

$\frac{n(n-1)}{2}$

$n(n-1)$

$2n$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3!$

ANSWER 6

2. $4!$

ANSWER 24

3. $5!$

ANSWER 120

4. $6!$

ANSWER 720

5. Arranging 5 books

ANSWER 120

6. First and second prize among 5

ANSWER 20

7. Two on duty from 5

ANSWER 10

Medium · the standard question

7 PROBLEMS

1. Gold, silver, bronze among 8

ANSWER 336

2. President and secretary from 12

ANSWER 132

3. Choosing 3 from 6

ANSWER 20

4. Handshakes among 10

ANSWER 45

5. Pairs of vertices of a hexagon

ANSWER 15

6. Diagonals of a hexagon

ANSWER 9

7. Seating 6 pupils in 6 chairs

ANSWER 720

Hard · stretch and reason

7 PROBLEMS

1. Handshakes among n people

ANSWER $\frac{n(n-1)}{2}$

2. Diagonals of an n -gon

ANSWER $\frac{n(n-3)}{2}$

3. Choosing 2 from 20

ANSWER 190

4. Arranging 4 of 7 books on a shelf

ANSWER 840

5. How many 3-letter “words” from $АБВГД$ with no repeats?

ANSWER 60

6. Why divide by $k!$ for a selection?

ANSWER Each group has been counted in all its orders

7. $10!$ is about

ANSWER 3.6 million

07 Homework

Homework — 5 tasks

Write “order matters” or “order does not matter” beside every question before counting.

1. In how many ways can 6 books be arranged on a shelf?
2. From 9 athletes, in how many ways can the first three places be awarded?
3. How many pairs on duty can be chosen from 8 pupils?
4. How many handshakes take place among 12 people?

5. Find the number of diagonals of an octagon.

Methods of solving combinatorial problems

Four working methods — the tree, the table, the rules, and counting the opposite case.

LESSON 161–163

3 × 40 MIN

MATEMATIKA 7, §67

S8 13

LESSON CLOCK

15'

30'

30'

30'

15'

Choosing a method	15 min
Dealing with a restriction first	30 min
Counting the opposite case	30 min
Worked problems	30 min
Checking an answer	15 min

LEARNING OBJECTIVES

- Choose a method that fits the size of the problem.
- Handle a restriction by dealing with it first.
- Count a difficult case as the total minus the opposite one.
- Check an answer by a second method or by a small case.

TEXTBOOK

National	pp. 484–494
Cambridge	Stage 8, pp. 126–134
Workbook	Exercise 13.3

01

Explanation

Choosing a method

METHOD	USE IT WHEN	DANGER
a systematic list	under about 20 outcomes	missing one
a tree diagram	two or three stages	the page runs out
the two rules	most problems	choosing the wrong rule
the opposite case	“at least one” problems	forgetting to subtract from the total

LIST SMALL, CALCULATE LARGE

A list is worth writing when it can be finished in a minute — and it is then the best check on a calculation. Beyond that, the rules are the only practical route.

Dealing with a restriction first

Whatever is restricted is chosen at the first stage; the free choices follow.

PROBLEM	FIRST	THEN	ANSWER
3-figure numbers from 1,2,3,4,5, no repeats	any digit: 5	$4 \cdot 3$	60
the same, but even	last digit 2 or 4: 2	$4 \cdot 3$	24
the same, but starting with 5	first digit fixed: 1	$4 \cdot 3$	12
seating 4 pupils with one fixed at the end	that pupil: 1	$3!$	6

CHOOSE THE RESTRICTED PLACE FIRST, NOT LAST

If the units digit is chosen last, the number of choices left depends on what went before and the product is no longer a single multiplication. Taking the restriction first keeps every stage independent.

Counting the opposite case

“At least one” is nearly always easier as “all of them minus none of them”.

at least one = total – none

PROBLEM	TOTAL	NONE	ANSWER
3-figure codes with at least one 7	1000	$9^3 = 729$	271
three coins with at least one head	8	1	7
two dice with at least one six	36	25	11

“AT LEAST ONE” IS NOT “EXACTLY ONE”

Two dice give 11 outcomes with at least one six, but only 10 with exactly one — the double six belongs to the first count and not to the second.

Worked problems

PROBLEM	WORKING	ANSWER
3-figure numbers from 1,2,3,4,5 without repeats	$5 \cdot 4 \cdot 3$	60
how many of them are even	$2 \cdot 4 \cdot 3$	24
how many are odd	$60 - 24$	36
how many are greater than 300	$3 \cdot 4 \cdot 3$	36
a committee of 2 from 4 boys and 3 girls with one of each	$4 \cdot 3$	12

EVEN AND ODD SHOULD ADD TO THE TOTAL

$24 + 36 = 60$ ✓. Splitting a count into two cases gives a free check: the parts must rebuild the whole.

Checking an answer

CHECK	HOW
a small case	redo the problem with 2 or 3 objects and list them
the complement	the case and its opposite must add to the total
a second method	tree against rules, or list against product
size	an answer larger than the total is impossible

THE SMALL CASE IS THE BEST CHECK THERE IS

If a formula gives 45 handshakes for 10 people, test it on 3 people: it should give 3, and three people shaking hands really do make three handshakes.

02 Worked examples

EXAMPLE 1 How many 3-figure numbers with different digits can be made from 1, 2, 3, 4, 5, and how many of them are even?

① $5 \cdot 4 \cdot 3 = 60$ altogether.

② Even: the units digit must be 2 or 4 — 2 ways.
The restriction first.

③ The other two places: $4 \cdot 3 = 12$.

④ $2 \cdot 12 = 24$ even numbers.
Check: $24 + 36 = 60$ ✓

ANSWER

60 in all, 24 even

EXAMPLE 2 How many 3-figure codes from 10 digits contain at least one 7?

① Total: $10^3 = 1000$.

② With no 7: $9^3 = 729$.
Nine digits remain at each place.

③ $1000 - 729$

④ $= 271$

ANSWER

271

EXAMPLE 3 A committee of two is chosen from 4 boys and 3 girls, one of each. How many committees?

① One boy: 4 ways.

② One girl: 3 ways.

③ “And” — multiply.

④ $4 \cdot 3 = 12$

ANSWER

12

03 Interactive model

Ask for the number of three-coin outcomes with at least one head; most of the class starts listing, and the total-minus-none route wins by a distance.

Restrictions and opposites

3-figure numbers from 1,2,3,4,5 without repeats:

15

60

125

120

How many are even?

12

24

30

36

How many are odd?

24

30

36

40

The restricted place should be chosen:

first

last

in the middle

at random

“At least one” is best counted as:

exactly one

total minus none

total plus none

none

3-figure codes with at least one 7:

271

300

729

100

Three coins with at least one head:

3

6

7

8

Two dice with at least one six:

6

10

11

12

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Method	Usul	Метод
Restriction	Cheklov	Ограничение
Complementary counting	Teskarisini sanash	Подсчёт дополнения
At least one	Kamida bitta	Хотя бы один
Systematic list	Tartibli ro‘yxat	Систематический перечень
To check	Tekshirmoq	Проверить
Small case	Kichik holat	Малый случай
Total	Umumiy soni	Общее число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A restriction should be handled:

first

last

never

twice

“At least one” equals:

exactly one

total minus none

none

all

3-figure codes with at least one 7 number:

271

729

100

300

Two dice with at least one six:

10

11

12

6

A good check on a count is:

doing it again the same way

a small case

guessing

rounding

Even and odd counts must:

be equal

add to the total

differ by one

multiply to the total

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 3-figure numbers from 1,2,3,4,5, no repeats

ANSWER 60

2. How many are even

ANSWER 24

3. How many are odd

ANSWER 36

4. How many start with 5

ANSWER 12

5. Three coins: outcomes in all

ANSWER 8

6. Three coins with at least one head

ANSWER 7

7. Two dice: outcomes in all

ANSWER 36

Medium · the standard question

7 PROBLEMS

1. 3-figure codes with at least one 7

ANSWER 271

2. Two dice with at least one six

ANSWER 11

3. Two dice with exactly one six

ANSWER 10

4. A committee of one boy and one girl from 4 and 3

ANSWER 12

5. 3-figure numbers from 1,2,3,4,5 greater than 300

ANSWER 36

6. Seating 4 pupils with one fixed at the end

ANSWER 6

7. 4-figure PINs with no 0

ANSWER 6561

Hard · stretch and reason

7 PROBLEMS

1. 4-figure codes with at least one 7

ANSWER 3439

2. Three dice with at least one six

ANSWER 91

3. 3-digit numbers with no repeated digit and even

ANSWER 328

4. Committees of 2 from 7 pupils

ANSWER 21

5. Committees of 2 from 7 with at least one of 3 girls

ANSWER 15

6. Why check with a small case?

ANSWER It can be listed in full and compared

7. Arrangements of 5 books with two particular ones together

ANSWER 48

07 Homework

Homework — 5 tasks

For every “at least one”, count the opposite case and subtract.

1. How many 3-figure numbers with different digits can be made from 2, 3, 4, 5, 6?
2. How many of them are even?
3. How many 4-figure codes from 10 digits contain at least one 9?
4. How many outcomes of four coins contain at least one tail?
5. A committee of one boy and one girl is chosen from 5 boys and 6 girls. How many committees?

Practical exercises

— experimental and theoretical probability

A Cambridge insert: what the die should do, what it actually does, and why the two converge.

LESSON 164

1 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 13

LESSON CLOCK

10'

12'

10'

6'

2'

Theoretical probability	10 min
The experiment	12 min
Why they differ	10 min
Expected numbers	6 min
Homework	2 min

LEARNING OBJECTIVES

- Calculate a theoretical probability by counting outcomes.
- Calculate an experimental probability from a frequency table.
- Explain why the two differ and what makes them close.
- Estimate the expected number of successes in a run of trials.

TEXTBOOK

National	pp. 495–498
Cambridge	Stage 8, pp. 126–134
Workbook	Exercise 13.4

01

Explanation

Theoretical probability

$$p = \frac{\text{favourable outcomes}}{\text{all outcomes}}$$

It is a counting problem — which is why combinatorics came first.

EVENT	FAVOURABLE	ALL	P
a six on one die	1	6	$\frac{1}{6} \approx 0.17$
an even number	3	6	$\frac{1}{2}$
a head on a coin	1	2	0.5
a total of 7 on two dice	6	36	$\frac{1}{6}$

EVERY PROBABILITY LIES BETWEEN 0 AND 1

An impossible event has $P = 0$, a certain one $P = 1$. An answer outside that range is a signal that the counting went wrong.

The experiment

Throw a die 60 times and tabulate. The **relative frequency** of a face is the number of times it appeared divided by the number of throws.

FACE	1	2	3	4	5	6
frequency	8	12	9	11	10	10
relative frequency	0.13	0.20	0.15	0.18	0.17	0.17

The theoretical value is $\frac{1}{6} \approx 0.17$ for every face; the experiment scatters around it.

THE FREQUENCIES MUST ADD TO THE NUMBER OF THROWS

$8 + 12 + 9 + 11 + 10 + 10 = 60 \checkmark$, and the relative frequencies add to 1. Both checks take a moment and catch a miscount.

Why they differ

THROWS	SIXES	RELATIVE FREQUENCY	DISTANCE FROM 0.17
10	3	0.30	far
60	10	0.17	close
600	104	0.173	closer

A LONG RUN OF SIXES DOES NOT MAKE THE NEXT SIX LESS LIKELY

The die has no memory. Relative frequency settles down because later throws swamp the early ones, not because the die corrects itself.

Expected numbers

expected number = $P \times$ number of trials

EXPERIMENT	WORKING	EXPECTED
sixes in 60 throws	$\frac{1}{6} \cdot 60$	10
heads in 50 tosses	$0.5 \cdot 50$	25
evens in 90 throws	$0.5 \cdot 90$	45
totals of 7 in 180 throws of two dice	$\frac{1}{6} \cdot 180$	30

“EXPECTED” MEANS ON AVERAGE, NOT EXACTLY

Ten sixes in sixty throws is the average outcome, not a promise. Getting eight or thirteen is ordinary; getting none at all would be evidence that the die is biased.

02 Worked examples

EXAMPLE 1 A die is thrown 60 times and a six appears 13 times. Compare the experimental and theoretical probabilities.

① Experimental: $\frac{13}{60} \approx 0.22$.

② Theoretical: $\frac{1}{6} \approx 0.17$.

③ The experimental value is a little higher.
Ordinary variation in 60 throws.

ANSWER

0.22 against 0.17

EXAMPLE 2 How many heads would you expect in 50 tosses of a fair coin?

① $P = 0.5$

② $0.5 \cdot 50$

③ $= 25$
On average, not exactly.

ANSWER

25

EXAMPLE 3 Two dice are thrown. Find the probability of a total of 7.

① All outcomes: $6 \cdot 6 = 36$.

② Favourable: $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$ — six of them.

③
$$P = \frac{6}{36} = \frac{1}{6}$$

ANSWER

$$\frac{1}{6}$$

03 **Interactive model**

Pool the whole class's throws on the board: each pupil's twenty look erratic, and the combined six hundred sit visibly close to a sixth.

Theoretical, experimental, expected

P (a six on one die) is:

$$\frac{1}{2}$$

$$\frac{1}{3}$$

$$\frac{1}{6}$$

$$\frac{1}{12}$$

P (an even number) is:

$$\frac{1}{6}$$

$$\frac{1}{3}$$

$$\frac{1}{2}$$

$$\frac{2}{3}$$

A six in 13 of 60 throws gives an experimental probability of:

0.13

0.17

0.22

0.60

Expected sixes in 60 throws:

6

10

13

30

Expected heads in 50 tosses:

10

25

50

5

Relative frequencies of all faces add to:

0

0.5

1

6

After five sixes in a row, the next throw gives a six with probability:

$$\frac{1}{6}$$

less than $\frac{1}{6}$

more than $\frac{1}{6}$

0

Experimental values approach theoretical ones when:

the die is heavier

more trials are made

fewer trials are made

never

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Probability	Ehtimollik	Вероятность
Theoretical	Nazariy	Теоретическая
Experimental	Tajribaviy	Экспериментальная
Relative frequency	Nisbiy chastota	Относительная частота
Trial	Sinov	Испытание
Fair	Bir jinsli	Правильный
Biased	Nosoz	Смещённый
Expected number	Kutilayotgan soni	Ожидаемое число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Theoretical probability is found by:

experiment

counting outcomes

guessing

measuring

Experimental probability is found by:

counting outcomes

dividing frequency by trials

multiplying

estimating

Every probability lies between:

0 and 1

-1 and 1

0 and 100

any values

Expected sixes in 180 throws:

18

30

36

60

$P(\text{total } 7 \text{ on two dice})$ is:

$\frac{1}{36}$

$\frac{1}{12}$

$\frac{1}{6}$

$\frac{1}{9}$

More trials make the experimental value:

further from

closer to

equal to

unrelated to

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $P(\text{a six on one die})$

ANSWER $\frac{1}{6}$

2. $P(\text{an even number})$

ANSWER $\frac{1}{2}$

3. $P(\text{a head})$

ANSWER 0.5

4. $P(\text{a number over 4})$

ANSWER $\frac{1}{3}$

5. Expected heads in 50 tosses

ANSWER 25

6. Expected sixes in 60 throws

ANSWER 10

7. Relative frequency of 12 in 60

ANSWER 0.20

Medium · the standard question

7 PROBLEMS

1. A six in 13 of 60 throws: the experimental probability

ANSWER ≈ 0.22

2. $P(\text{total } 7 \text{ on two dice})$

ANSWER $\frac{1}{6}$

3. $P(\text{total } 12 \text{ on two dice})$

ANSWER $\frac{1}{36}$

4. Expected totals of 7 in 180 throws

ANSWER 30

5. Do relative frequencies add to 1?

ANSWER Yes

6. After five sixes, $P(\text{six})$ is

ANSWER $\frac{1}{6}$

7. Expected evens in 90 throws

ANSWER 45

Hard · stretch and reason

7 PROBLEMS

1. $P(\text{at least one six on two dice})$

ANSWER $\frac{11}{36}$

2. $P(\text{exactly one six on two dice})$

ANSWER $\frac{10}{36} = \frac{5}{18}$

3. $P(\text{at least one head on three coins})$

ANSWER $\frac{7}{8}$

4. A die gives 40 sixes in 60 throws: what would you conclude?

ANSWER It is very likely biased

5. Expected sixes in 600 throws

ANSWER 100

6. Why does relative frequency settle down?

ANSWER Later trials outweigh the early variation

7. $P(\text{a total under 5 on two dice})$

ANSWER $\frac{6}{36} = \frac{1}{6}$

07 Homework

Homework — 5 tasks

Record every throw as you make it; filling the table afterwards from memory is not an experiment.

1. Throw a die 60 times, tabulate the results and find each relative frequency.
2. Compare your six values with $\frac{1}{6}$ and comment.
3. Find $P(\text{a total of } 7)$ and $P(\text{a total of } 2)$ for two dice.

4. How many heads would you expect in 80 tosses of a coin?
5. Explain why a run of five heads does not change the probability of the next toss.

Practical exercises — data collection and sampling

A Cambridge insert: what to ask, whom to ask, and how a bad sample ruins a good question.

LESSON 165

1 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 6.1–6.2

LESSON CLOCK

8'

12'

12'

6'

2'

Population and sample	8 min
When a sample misleads	12 min
Writing the question	12 min
Tally and frequency	6 min
Homework	2 min

LEARNING OBJECTIVES

- Distinguish a population from a sample.
- Explain why a sample must be representative, and how bias creeps in.
- Write clear survey questions without leading or overlapping options.
- Record data in a tally and frequency table.

TEXTBOOK

National	pp. 499–502
Cambridge	Stage 8, pp. 58–65
Workbook	Exercise 6.1

01

Explanation

Population and sample

TERM	MEANING	EXAMPLE
population	everybody the question is about	all 800 pupils in the school
sample	the part actually asked	50
census	asking the whole population	all 800

WHY SAMPLE AT ALL

A census is exact but slow and expensive. A well-chosen sample of 50 can describe 800 pupils well enough to act on — which is the trade every survey makes.

When a sample misleads

QUESTION	SAMPLE TAKEN	WHAT GOES WRONG
"How much sport does a pupil do?"	the football team	far too much sport
"What is the favourite subject?"	one maths class	maths over-represented
"How do pupils travel to school?"	those in the bus queue	walkers excluded
"Should lessons start later?"	pupils arriving late	the keenest to say yes

A BIG SAMPLE DOES NOT FIX A BIASED ONE

Asking the whole football team instead of half of it makes the wrong answer more confident, not more correct. Bias is repaired by *how* the sample is chosen, not by its size.

Writing the question

FAULT	BAD	BETTER
leading	"Don't you agree maths is the best subject?"	"Which subject do you like most?"
overlapping options	0–5, 5–10	0–4, 5–9
gaps in the options	0–4, 6–9	0–4, 5–9
vague	"Do you read often?"	"How many books did you read last month?"
two questions in one	"Do you like maths and science?"	ask them separately

OPTIONS MUST COVER EVERYTHING, ONCE

Every answer must fit exactly one box: no overlaps, no gaps. That single rule fixes most faulty questionnaires.

Tally and frequency

BOOKS READ LAST MONTH	TALLY	FREQUENCY
0		7
1		13
2		10
3		6
4 or more		4

The frequencies add to 40 — the size of the sample, which is always worth checking.

TALLY IN FIVES

Four strokes and a cross-stroke make groups that can be counted at a glance. It is a small habit that removes most counting errors from a class survey.

02

Worked examples

EXAMPLE 1 A school has 800 pupils. A survey asks 50 of them. Name the population and the sample.

1

The question is about all the pupils.

2

Population: the 800.

3

Sample: the 50 asked.

ANSWER

Population 800, sample 50

EXAMPLE 2 Why is the football team a poor sample for “how much sport does a pupil do?”

- ① Team members were chosen because they play sport.
- ② Their hours are far above average.
- ③ The sample is biased and overstates the answer.

ANSWER

It is biased

EXAMPLE 3 What is wrong with the options 0–5, 5–10, 10–15?

- ① A pupil with exactly 5 fits two boxes.
- ② The same for 10.
- ③ Use 0–4, 5–9, 10–14.

ANSWER

The options overlap

03 Interactive model

Run one badly worded question on the class and one careful version of the same question; the difference in the answers makes the lesson better than any list of rules.

Good sample, good question?

Everybody the question is about is the:

sample

population

tally

frequency

The part actually asked is the:

sample

population

census

bias

Asking the football team about sport gives:

a fair answer

a biased answer

no answer

a census

A bigger biased sample is:

less biased

just as biased

unbiased

a census

“Don’t you agree maths is best?” is:

clear

leading

vague

fine

Options 0–5 and 5–10 are:

fine

overlapping

gapped

leading

Options 0–4 and 6–9 have:

an overlap

a gap

nothing wrong

too many boxes

Frequencies should add to:

1

the sample size

100

the population

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Population	Bosh to'plam	Генеральная совокупность
Sample	Tanlanma	Выборка
Representative	Vakillik qiluvchi	Репрезентативная
Bias	Siljish	Смещение
Random	Tasodifiy	Случайный
Questionnaire	So'rovnoma	Анкета
Tally	Sanoq belgisi	Отметка
Frequency	Chastota	Частота

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A sample is:

everybody

a part of the population

a question

a tally

A census asks:

a sample

the whole population

nobody

ten people

Bias is cured by:

a bigger sample

a better chosen sample

more questions

rounding

A leading question:

is clear

suggests an answer

has options

is short

Options must be:

overlapping

without gaps or overlaps

few

wide

A tally is grouped in:

twos

threes

fives

tens

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. All 800 pupils are the

ANSWER population

2. The 50 asked are the

ANSWER sample

3. Asking everybody is a

ANSWER census

4. A sample chosen badly gives

ANSWER bias

5. “Don’t you agree maths is best?” is

ANSWER a leading question

6. 0–5 and 5–10 are

ANSWER overlapping

7. Frequencies should add to

ANSWER the sample size

Medium · the standard question

7 PROBLEMS

1. Why is the football team a poor sample for sport?

ANSWER They were chosen because they play sport

2. Why is the bus queue a poor sample for travel?

ANSWER Walkers are excluded

3. Fix the options 0–5, 5–10

ANSWER 0–4, 5–9

4. Fix the options 0–4, 6–9

ANSWER 0–4, 5–9

5. Rewrite “Do you read often?”

ANSWER “How many books did you read last month?”

6. Does doubling a biased sample help?

ANSWER No

7. Frequencies 7, 13, 10, 6, 4: the sample size

ANSWER 40

Hard · stretch and reason

7 PROBLEMS

1. Design a fair way to choose 50 from 800 pupils

ANSWER Number them and draw 50 at random

2. Why not survey only your friends?

ANSWER They are not representative of the school

3. What is wrong with “Do you like maths and science?”

ANSWER It asks two questions at once

4. A survey on school lunches asks pupils leaving the canteen: the fault

ANSWER Those who never eat there are excluded

5. Why must options have no gaps?

ANSWER Some answers would fit no box

6. A sample of 50 from 800 is what fraction?

ANSWER $\frac{1}{16}$

7. When is a census worth doing?

ANSWER When the population is small or the answer must be exact

07 Homework

Homework — 5 tasks

Write your question out fully before asking anybody anything.

1. Write a survey question with five non-overlapping options about time spent reading.
2. Name the population and describe a fair sample for “how do pupils travel to school?”
3. Explain one way a sample can be biased and how to avoid it.
4. Collect 20 answers to your question and record them in a tally table.

5. Find the frequency of each option and check that they add to 20.

Practical exercises

— interpreting and discussing results

A Cambridge insert: averages, spread, and the graph that tells the truth about the data.

LESSON 166

1 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 16

LESSON CLOCK

10'

12'

10'

6'

2'

The three averages	10 min
Which average to use	12 min
Comparing two sets	10 min
Misleading graphs	6 min
Homework	2 min

LEARNING OBJECTIVES

- Find the mean, median, mode and range of a small data set.
- Choose the average that describes the data honestly.
- Read a bar chart and a frequency table and compare two sets.
- Recognise a misleading graph.

TEXTBOOK

National	pp. 503–506
Cambridge	Stage 8, pp. 158–166
Workbook	Exercise 16.1

01

Explanation

The three averages

MEASURE	HOW	FOR 3, 5, 5, 6, 8, 9, 12
mean	add and divide by the count	$\frac{48}{7} \approx 6.9$
median	the middle value in order	6
mode	the commonest value	5
range	largest minus smallest	9

ORDER THE DATA BEFORE TAKING A MEDIAN

The middle of the list as written is not the median unless the list is sorted. With an even number of values, average the two middle ones.

Which average to use

DATA	BEST AVERAGE	WHY
test marks, no extremes	mean	it uses every value
salaries with one very large one	median	the outlier drags the mean up
shoe sizes sold	mode	the shop wants the commonest
colours of cars	mode only	they cannot be added or ordered

Example: 2, 3, 3, 4, 88 has mean 20 and median 3. The mean describes no one in the list.

THE RANGE IS NOT AN AVERAGE

It measures spread, not the centre. Two classes can share a mean of 60 with ranges of 10 and 70 — very different classes, and the range is what says so.

Comparing two sets

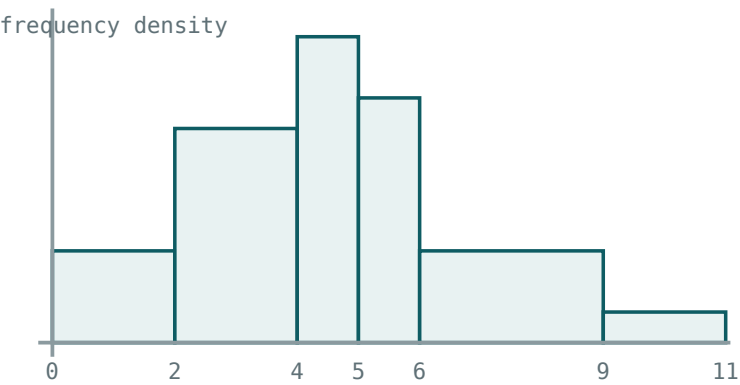
CLASS	MEAN	RANGE	WHAT IT MEANS
7A	62	14	similar marks, tightly grouped
7B	62	55	the same average, wildly spread

A fair comparison quotes both: one measure of centre and one of spread. Quoting the mean alone would make the two classes look identical.

TWO SENTENCES FINISH ANY COMPARISON

“On average 7B did as well as 7A; but 7B’s marks were far more spread out.” Centre, then spread — and both in words, not just numbers.

Misleading graphs



with unequal widths the AREA is the frequency

A bar chart, drawn honestly

TRICK	WHAT IT DOES	THE FIX
vertical axis not starting at 0	small differences look huge	start at 0, or mark the break
bars of different widths	area misleads the eye	equal widths
no scale on an axis	nothing can be checked	label both axes
pictures scaled in two directions	doubling looks like quadrupling	use bars

ALWAYS READ THE VERTICAL AXIS FIRST

A bar twice as tall as another means twice as much only if the axis starts at zero. This is the commonest way an honest set of numbers is used to tell a lie.

02 Worked examples

EXAMPLE 1 Find the mean, median, mode and range of 3, 5, 5, 6, 8, 9, 12.

- 1 Sum = 48, count 7, so the mean is ≈ 6.9 .
- 2 Already in order; the middle value is 6.
- 3 5 appears twice — it is the mode.
- 4 Range = $12 - 3 = 9$.

ANSWER

Mean ≈ 6.9 , median 6, mode 5, range 9

EXAMPLE 2 Which average best describes 2, 3, 3, 4, 88?

- 1 Mean = $\frac{100}{5} = 20$.
- 2 Median = 3.
- 3 The 88 is an outlier and drags the mean far above every other value.
- 4 The median describes the data honestly.

ANSWER

The median

EXAMPLE 3 Two classes both have mean 62; the ranges are 14 and 55. What can be said?

- ① On average they performed equally.
- ② The second class is far more spread out.
- ③ Some pupils there did much better and some much worse.

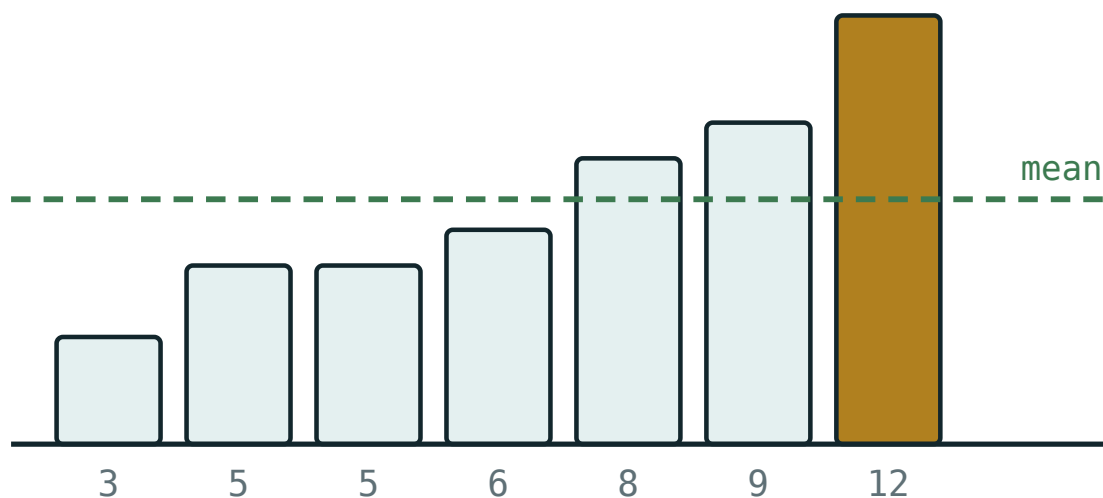
ANSWER

Same centre, very different spread

03 Interactive model

Show the same bar chart twice, once with the axis from 0 and once from 55; the class sees the same data tell two different stories.

Change one value and watch the three averages



values **3, 5, 5, 6, 8, 9, 12**

mean **6.86**

median **6**

mode **5**

range **9**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Mean	O'rta arifmetik	Среднее арифметическое
Median	Mediana	Медиана
Mode	Moda	Мода
Range	Ayirma (tarqoqlik)	Размах
Outlier	Chetlangan qiymat	Выброс
Bar chart	Ustunli diagramma	Столбчатая диаграмма
To compare	Taqqoslash	Сравнить
Misleading	Chalg'ituvchi	Вводящий в заблуждение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The mean of 3, 5, 5, 6, 8, 9, 12 is about:

Its median is:

Its mode is:

Its range is:

3

6

9

12

With one huge outlier, the best average is the:

mean

median

mode

range

A bar chart whose axis starts at 55 is:

fine

misleading

impossible

a histogram

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The mean of 4, 6, 8

ANSWER 6

2. The median of 4, 6, 8

ANSWER 6

3. The mode of 4, 6, 6, 8

ANSWER 6

4. The range of 4, 6, 8

ANSWER 4

5. The mean of 3, 5, 5, 6, 8, 9, 12

ANSWER ≈ 6.9

6. Its median

ANSWER 6

7. Its range

ANSWER 9

Medium · the standard question

7 PROBLEMS

1. The mean of 2, 3, 3, 4, 88

ANSWER 20

2. Its median

ANSWER 3

3. Which average describes it better?

ANSWER The median

4. The median of 4, 7, 9, 12

ANSWER 8

5. Two classes, mean 62, ranges 14 and 55

ANSWER Same centre, different spread

6. Which average suits shoe sizes sold?

ANSWER The mode

7. Which suits car colours?

ANSWER The mode only

Hard · stretch and reason

7 PROBLEMS

1. Five numbers have mean 10: their sum

ANSWER 50

2. Four numbers have mean 7; a fifth is added and the mean becomes 8: the fifth number

ANSWER 12

3. A set of 5 numbers with mean 6, median 5 and mode 5

ANSWER e.g. 3, 5, 5, 7, 10

4. Why can the mean lie outside the commonest values?

ANSWER One extreme value pulls it away

5. Two sets with the same mean but ranges 2 and 40

ANSWER One is consistent, the other erratic

6. Why must a bar chart axis start at 0?

ANSWER Otherwise bar heights are not proportional

7. The mean of the first 10 whole numbers 1 to 10

ANSWER 5.5

07 Homework

Homework — 5 tasks

Order the data first; the median depends on it and the range does too.

- Find the mean, median, mode and range of 7, 4, 9, 4, 6, 10, 4.
- For 1, 2, 2, 3, 92, find the mean and median and say which describes the data better.
- Two classes have mean 55 with ranges 12 and 48. Write two sentences comparing them.
- Draw a bar chart of your own data with the axis starting at 0.
- Describe two ways a graph can mislead.

Practical exercises — ratio and proportion

A Cambridge insert: sharing in a ratio, scaling a recipe, and telling direct from inverse.

LESSON 167

1 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 12

LESSON CLOCK

8'

12'

12'

6'

2'

Ratios and their parts	8 min
Sharing in a ratio	12 min
The unitary method	12 min
Direct and inverse	6 min
Homework	2 min

LEARNING OBJECTIVES

- Simplify a ratio and share a quantity in a given ratio.
- Use the unitary method to scale a recipe or a price.
- Distinguish direct from inverse proportion.
- Use a scale on a map or a plan.

TEXTBOOK

National	pp. 507–510
Cambridge	Stage 8, pp. 118–125
Workbook	Exercise 12.2

01

Explanation

Ratios and their parts

A ratio compares quantities of the same kind. It simplifies like a fraction, by dividing every part by their common factor.

RATIO	SIMPLIFIED	PARTS IN ALL
12 : 18	2 : 3	5
30 : 45 : 75	2 : 3 : 5	10
0.5 : 2	1 : 4	5
$\frac{1}{2} : \frac{1}{3}$	3 : 2	5

BOTH QUANTITIES MUST BE IN THE SAME UNITS

50 cm : 2 m is not 50 : 2. Convert first: 50 : 200 = 1 : 4.

Sharing in a ratio

SHARING 120 IN THE RATIO 2:3		
STEP		
add the parts	$2 + 3 = 5$	
find one part	$120 \div 5 = 24$	
multiply out	$2 \cdot 24 = 48, 3 \cdot 24 = 72$	
check	$48 + 72 = 120 \checkmark$	

AMOUNT	RATIO	SHARES
120	2 : 3	48, 72
210	3 : 4	90, 120
300	1 : 2 : 3	50, 100, 150

THE SHARES MUST REBUILD THE TOTAL

Adding them back is a two-second check that catches nearly every arithmetic slip in this topic.

The unitary method

Find the value of one unit, then multiply up.

PROBLEM	ONE UNIT	ANSWER
5 pens cost 15 000: 8 pens	3000	24 000
a recipe for 4 needs 600 g flour: for 6	150 g	900 g
3 workers lay 90 m in a day: 7 workers	30 m	210 m

SCALES ARE RATIOS TOO

A map at 1 : 100 000 shows 1 cm for every 1 km on the ground, so 7 cm between two towns means 7 km.
Read the scale as “one of these is that many of those”.

Direct and inverse

KIND	AS ONE GROWS	CONSTANT	EXAMPLE
direct	the other grows	$\frac{y}{x}$	cost of n identical pens
inverse	the other shrinks	xy	workers and the days a job takes

Example: 4 workers finish a job in 6 days; how long would 8 take? The product $4 \cdot 6 = 24$ worker-days is fixed, so 8 workers need 3 days.

NOT EVERYTHING THAT CHANGES TOGETHER IS PROPORTIONAL

Doubling the workers may not halve the time if they get in each other's way, and a child twice as old is not twice as tall. Proportion is a model, and it has to be checked against sense.

02

Worked examples

EXAMPLE 1 Share 120 in the ratio 2 : 3.

1

2 + 3 = 5 parts.

2

120 ÷ 5 = 24 for one part.

3

48 and 72.
Check: 48 + 72 = 120 ✓

ANSWER

48 and 72

EXAMPLE 2 A recipe for 4 people needs 600 g of flour. How much for 6?

① One person: $600 \div 4 = 150 \text{ g}$.

② Six people: $150 \cdot 6$.

③ $= 900 \text{ g}$

ANSWER

900 g

EXAMPLE 3 4 workers finish a job in 6 days. How long would 8 workers take?

① More workers, less time — inverse proportion.

② $4 \cdot 6 = 24$ worker-days.

③ $24 \div 8 = 3$

④ Three days.

ANSWER

3 days

03 Interactive model

Scale a real recipe from four people to the size of the class; the arithmetic is identical to the textbook exercise and nobody asks what it is for.

Direct, inverse, or neither?

12 : 18 simplifies to:

2 : 3

3 : 2

6 : 9

1 : 2

50 cm : 2 m is:

50 : 2

1 : 4

25 : 1

1 : 40

120 in the ratio *2 : 3* gives:

40, 80

48, 72

50, 70

60, 60

5 pens cost 15 000; 8 cost:

20 000

24 000

30 000

18 000

Cost against number of identical pens is:

direct

inverse

neither

constant

Workers against days for a fixed job is:

direct

inverse

neither

constant

4 workers take 6 days; 8 take:

12

3

2

4

A map at 1 : 100 000: 7 cm means:

700 m

7 km

70 km

7 m

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Ratio	Nisbat	Отношение
Proportion	Proporsiya	Пропорция
To share	Taqsimlash	Разделить
Unitary method	Birlik usuli	Метод приведения к единице
Direct proportion	To'g'ri proporsionallik	Прямая пропорциональность
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Scale	Masshtab	Масштаб
Equivalent ratios	Teng nisbatlar	Равные отношения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$30 : 45 : 75$ simplifies to:

$2 : 3 : 5$

$1 : 2 : 3$

$3 : 4 : 5$

$6 : 9 : 15$

Sharing needs first:

the total

the sum of the parts

one share

the ratio reversed

210 in the ratio $3 : 4$ gives:

$90, 120$

$100, 110$

$70, 140$

$105, 105$

The unitary method finds:

the total

the value of one unit

the ratio

the difference

In inverse proportion the constant is:

$$\frac{y}{x}$$

xy

$x + y$

$x - y$

A recipe for 4 scaled to 6 multiplies amounts by:

1.5

2

0.5

6

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify $12 : 18$

ANSWER $2 : 3$

2. Simplify $30 : 45 : 75$

ANSWER $2 : 3 : 5$

3. Simplify $50\text{ cm} : 2\text{ m}$

ANSWER $1 : 4$

4. Share 120 in $2 : 3$

ANSWER 48 and 72

5. Share 210 in $3 : 4$

ANSWER 90 and 120

6. 5 pens cost 15 000: one pen

ANSWER 3000

7. So 8 pens cost

ANSWER 24 000

Medium · the standard question

7 PROBLEMS

1. Share 300 in 1 : 2 : 3

ANSWER 50, 100, 150

2. A recipe for 4 needs 600 g: for 6

ANSWER 900 g

3. 3 workers lay 90 m a day: 7 workers

ANSWER 210 m

4. 4 workers take 6 days: 8 workers

ANSWER 3 days

5. A map at 1 : 100 000: 7 cm on it

ANSWER 7 km

6. Simplify $\frac{1}{2} : \frac{1}{3}$

ANSWER 3 : 2

7. Cost against number of pens is

ANSWER direct proportion

Hard · stretch and reason

7 PROBLEMS

1. Share 480 in 3 : 5 : 8

ANSWER 90, 150, 240

2. 6 taps fill a tank in 4 hours: 8 taps

ANSWER 3 hours

3. Two shares in ratio 3 : 5 differ by 40: the total

ANSWER 160

4. A map at 1 : 25 000: 3 km on the ground

ANSWER 12 cm

5. Is a child's height directly proportional to age?

ANSWER No — growth is not steady

6. If y is inversely proportional to x and $y = 12$ when $x = 2$, find y when $x = 8$

ANSWER 3

7. A recipe for 4 scaled to 10 multiplies by

ANSWER 2.5

07 Homework

Homework — 5 tasks

Convert to the same units before simplifying any ratio.

1. Simplify 24 : 36, 75 cm : 3 m and 20 : 30 : 50.
2. Share 360 in the ratio 4 : 5.
3. 7 books cost 49 000 sum. Find the cost of 11.
4. 5 workers take 12 days. How long would 10 take?
5. A map at 1 : 50 000 shows two towns 9 cm apart. Find the real distance.

Practical exercises

— surface area and volume of prisms

A Cambridge insert: base area times height for the volume, and the net for the surface.

LESSON 168

1 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 15

LESSON CLOCK

8'

12'

12'

6'

2'

What a prism is	8 min
Volume	12 min
Surface area from the net	12 min
Units and capacity	6 min
Homework	2 min

LEARNING OBJECTIVES

- Use $V = Bh$ for any prism.
- Find the surface area from the net, as $2B + Ph$.
- Work with the units of volume and capacity.
- Solve practical problems about boxes, tanks and channels.

TEXTBOOK

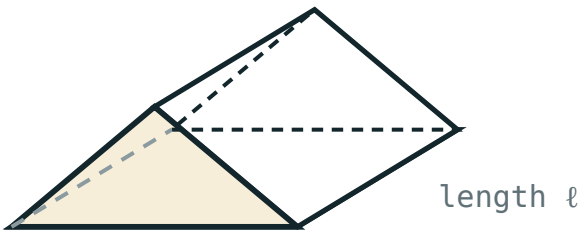
National	pp. 511–514
Cambridge	Stage 8, pp. 148–157
Workbook	Exercise 15.3

01

Explanation

What a prism is

A **prism** has two identical parallel faces — the bases — joined by rectangles. Its cross-section is the same all the way along.



base area S

$$V = S \cdot \ell$$

A prism: the base swept along its height

SOLID	A PRISM?	BASE
cuboid	yes	a rectangle
cube	yes	a square
triangular prism	yes	a triangle
cylinder	yes, in effect	a circle
pyramid	no	it narrows to a point

ONE FORMULA FOR ALL OF THEM

Because the cross-section never changes, the volume is the base area repeated all the way up. That is $V = Bh$, whatever shape the base has.

Volume

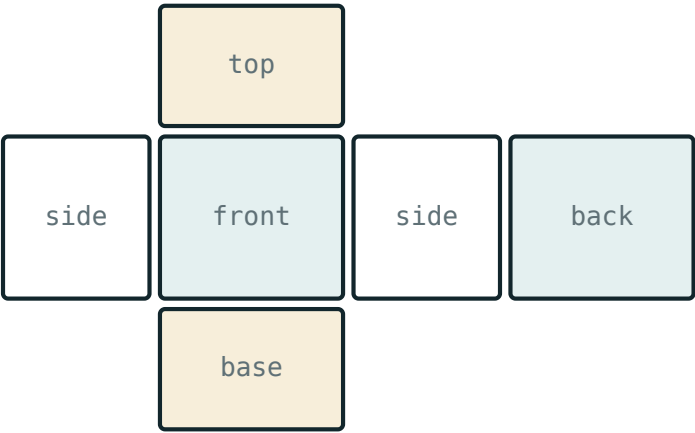
$$V = Bh$$

PRISM	BASE AREA B	HEIGHT H	VOLUME
cuboid $5 \times 4 \times 3$	20	3	60 cm^3
cube of edge 4	16	4	64 cm^3
triangular, base 6, height 4, length 10	12	10	120 cm^3
cylinder $r = 7, h = 10$	154	10	1540 cm^3

THE HEIGHT IS MEASURED PERPENDICULAR TO THE BASE

For a triangular prism lying on its side, the “height” of the prism is its length, and the triangle's own height belongs to the base area. Two different heights appear, and confusing them is the usual error.

Surface area from the net



surface area = the area of the whole net

The net of a prism: two bases and one long rectangle

$$S = 2B + Ph$$

where P is the perimeter of the base: the two bases, plus one rectangle whose width is the base's perimeter and whose height is the prism's.

PRISM	$2B$	Ph	S
cuboid $5 \times 4 \times 3$	40	$18 \cdot 3 = 54$	94 cm^2
cube of edge 4	32	$16 \cdot 4 = 64$	96 cm^2
triangular $3, 4, 5$ base, length 10	12	$12 \cdot 10 = 120$	132 cm^2

A CUBE'S SURFACE IS $6A^2$

Six identical faces — and $2B + Ph$ agrees: $2a^2 + 4a \cdot a = 6a^2$. A formula that survives a special case is a formula worth trusting.

Units and capacity

CONVERSION	FACTOR	WHY
$1\text{ cm}^3 = 1000\text{ mm}^3$	10^3	three directions
$1\text{ litre} = 1000\text{ cm}^3$	10^3	a 10 cm cube
$1\text{ m}^3 = 1000\text{ litres}$	$100^3 \div 1000$	$1\,000\,000\text{ cm}^3$

PROBLEM	WORKING	ANSWER
a tank $50 \times 40 \times 30$ cm: its capacity	$60\,000\text{ cm}^3$	60 litres
a box $20 \times 20 \times 20$: its volume	8000 cm^3	8 litres
a pool $25 \times 10 \times 2$ m	500 m^3	500 000 litres

LENGTHS SCALE BY k , AREAS BY k^2 , VOLUMES BY k^3

Doubling every edge of a box multiplies its surface by 4 and its volume by 8. This is why a large tank holds far more water per square metre of metal than a small one.

02

Worked examples

EXAMPLE 1 Find the volume and surface area of a cuboid $5 \times 4 \times 3$ cm.

1

$B = 5 \cdot 4 = 20\text{ cm}^2, h = 3.$

2

$V = 60\text{ cm}^3$

3

$P = 2(5 + 4) = 18\text{ cm}, \text{ so } Ph = 54.$

4

$S = 2 \cdot 20 + 54 = 94\text{ cm}^2$

ANSWER

60 cm^3 and 94 cm^2

EXAMPLE 2 A triangular prism has a base triangle with sides 3, 4, 5 cm and length 10 cm. Find its volume and surface area.

① The triangle is right-angled: $B = \frac{1}{2} \cdot 3 \cdot 4 = 6 \text{ cm}^2$.

② $V = 6 \cdot 10 = 60 \text{ cm}^3$

③ $P = 3 + 4 + 5 = 12$, so $Ph = 120$.

④ $S = 12 + 120 = 132 \text{ cm}^2$

ANSWER

60 cm^3 and 132 cm^2

EXAMPLE 3 A tank measures $50 \times 40 \times 30 \text{ cm}$. How many litres does it hold?

① $V = 50 \cdot 40 \cdot 30$

② $= 60\,000 \text{ cm}^3$

③ $60\,000 \div 1000$

④ $= 60 \text{ litres}$.

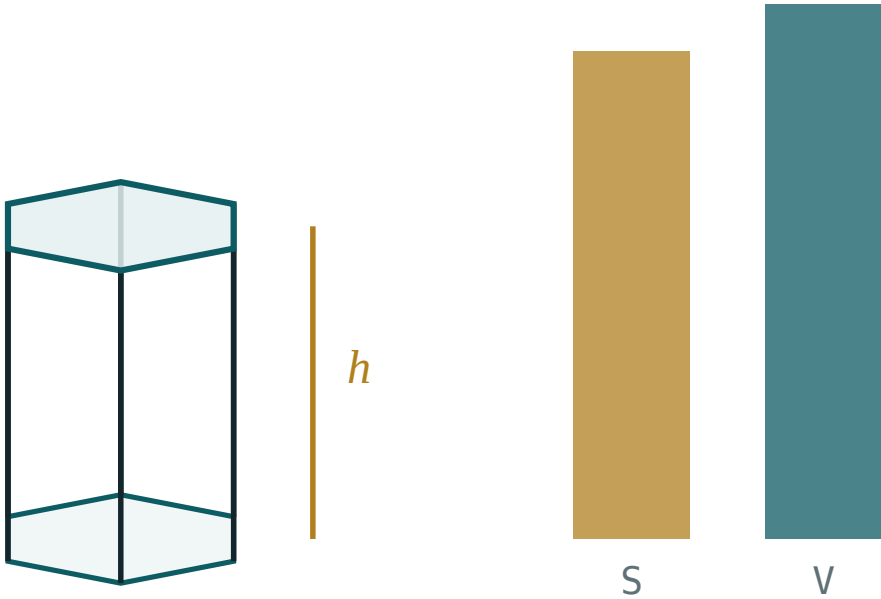
ANSWER

60 litres

03 Interactive model

Cut open a cardboard box along its edges and lay it flat; the surface-area formula is then read straight off the net rather than memorised.

Base area and height decide the volume



base area B **41.6**

perimeter P **24**

total S **227.1**

volume V **249.4**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Prism	Prizma	Призма
Base area	Asos yuzasi	Площадь основания
Height	Balandlik	Высота
Volume	Hajm	Объём
Surface area	To'la sirt	Площадь поверхности
Net	Yoyilma	Развёртка
Capacity	Sig'im	Вместимость
Litre	Litr	Литр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The volume of any prism is:

Bh

$\frac{1}{3} Bh$

$2B + Ph$

Ph

Its surface area is:

Bh

$2B + Ph$

$6a^2$

$2Ph$

A cuboid $5 \times 4 \times 3$ has volume:

12

20

60

94

And surface area:

60

94

96

120

1 litre equals:

100 cm³

1000 cm³

10 000 cm³

1 000 000 cm³

Doubling every edge multiplies the volume by:

2

4

6

8

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A cuboid $5 \times 4 \times 3$: its volume

ANSWER 60 cm^3

2. Its surface area

ANSWER 94 cm^2

3. A cube of edge 4: its volume

ANSWER 64 cm^3

4. Its surface area

ANSWER 96 cm^2

5. A cube of edge a : its surface area

ANSWER $6a^2$

6. 1 litre in cm^3

ANSWER 1000

7. 1 m^3 in litres

ANSWER 1000

Medium · the standard question

7 PROBLEMS

1. A triangular prism, base 6, height 4, length 10

ANSWER 120 cm^3

2. A 3, 4, 5 prism of length 10: its surface area

ANSWER 132 cm^2

3. A cylinder $r = 7$, $h = 10$ ($\pi = \frac{22}{7}$)

ANSWER 1540 cm^3

4. A tank $50 \times 40 \times 30 \text{ cm}$: its capacity

ANSWER 60 litres

5. A box $20 \times 20 \times 20$: its capacity

ANSWER 8 litres

6. A pool $25 \times 10 \times 2 \text{ m}$

ANSWER 500 m^3

7. That pool in litres

ANSWER 500 000

Hard · stretch and reason

7 PROBLEMS

1. Doubling every edge multiplies the surface by

ANSWER 4

2. And the volume by

ANSWER 8

3. A cuboid of volume 120 and base 5×4 : its height

ANSWER 6

4. A cube of volume 125 cm^3 : its surface area

ANSWER 150 cm^2

5. A cube of surface 54 cm^2 : its volume

ANSWER 27 cm^3

6. A channel with trapezium cross-section $B = 15 \text{ cm}^2$ and length 4 m

ANSWER 6000 cm^3

7. Why does a big tank need less metal per litre?

ANSWER Surface grows as k^2 but volume as k^3

07 Homework

Homework — 5 tasks

cm^3 for a volume, cm^2 for a surface — write the units every time.

- Find the volume and surface area of a cuboid $8 \times 5 \times 4 \text{ cm}$.
- Find the volume and surface area of a cube of edge 6 cm.
- A triangular prism has base 8 cm, triangle height 5 cm and length 12 cm. Find its volume.
- A tank measures $80 \times 50 \times 40 \text{ cm}$. How many litres does it hold?
- A cube has volume 216 cm^3 . Find its edge and its surface area.

Annual control work, and the Grade 8 preview

The whole year in one paper — and what waits in Grade 8.

LESSON 169–170

2 × 40 MIN

MATEMATIKA 7, YILLIK NAZORAT ISHI

S8 REVIEW

LESSON CLOCK

3

45'

12'

15'

5'

Instructions	3 min
The paper	45 min
Answers	12 min
Diagnosis	15 min
Grade 8	5 min

LEARNING OBJECTIVES

- Work accurately across the whole year under time.
- Choose the right method without being told which chapter a question comes from.
- Diagnose the year's remaining weaknesses.
- See how each block of Grade 7 continues in Grade 8.

TEXTBOOK

National	the whole textbook
Cambridge	Stage 8, whole book
Workbook	Annual paper

01

Explanation

The paper — 40 marks, 45 minutes

Q	TASK	MARKS	QUARTER
1	Compute $-7 + 12 - (-5)$ and $\frac{2}{3} - \frac{1}{4}$	4	I
2	Expand $(3x - 2)^2$ and factorise $x^2 - 49$	5	II–III
3	Simplify $\frac{x^2 - 9}{x + 3}$ and state the excluded value	4	III
4	Solve $3(x - 2) = 2x + 5$	4	III
5	Two parallel lines are cut by a transversal; one angle is 115° : find the other seven	5	III
6	The angles of a triangle are in the ratio $2 : 3 : 4$: find them	4	III
7	Draw $y = 2x - 4$ and give both intercepts	5	IV
8	Solve $\{ x + y = 9 ; x - y = 3 \}$	5	IV
9	How many 3-figure codes from 10 digits have all digits different?	4	IV

NINE QUESTIONS, FOUR QUARTERS

Nothing is asked that has not been taught, and nothing is labelled with its chapter. Recognising which method a question wants is itself the skill being tested.

The answers

Q	ANSWER	THE MARK MOST OFTEN LOST
1	10	the double minus, and the common denominator
2	$9x^2 - 12x + 4$	the middle term, and squaring the 3
3	$x - 3$	$x \neq -3$
4	$x = 11$	expanding the bracket
5	115°	naming which pair is which
6	$40^\circ, 60^\circ, 80^\circ$	$9x = 180$
7	$(0; -4)$	using a ruler
8	$(6; 3)$	reporting both numbers
9	720	$10 \cdot 9 \cdot 8$

Diagnosis

IF YOU LOST MARKS ON	REVISE
Q1	integers and rational numbers — Quarter I
Q2	the abridged multiplication formulae — L98–109
Q3	algebraic fractions — L112–119
Q4	linear equations — L123–126
Q5	angles on parallel lines — L82–85
Q6	the angle sum of a triangle — L86–87
Q7	the linear function — L141–142
Q8	systems — L147–149
Q9	combinatorics — L155–163

REVISE THE BLOCK, NOT THE QUESTION

A lost mark is a symptom. Reworking one question fixes that question; rereading the block fixes the next twenty.

What Grade 8 expects

GRADE 7 BLOCK	GRADE 8 CONTINUES WITH
rational numbers	irrational numbers and square roots
abridged formulae	quadratic equations and completing the square
algebraic fractions	rational expressions and fractional equations
linear equations	inequalities and their intervals
parallel lines and triangles	quadrilaterals, Pythagoras and trigonometry
the linear function	$y = x^2$
systems	systems with a quadratic equation in them
combinatorics	probability from counted outcomes

NOTHING IS DROPPED

Every block above continues; none of it is replaced. The year's work is the ground Grade 8 stands on, which is why the diagnosis above is worth acting on over the summer.

02 Worked examples

EXAMPLE 1 Model answer, Q4: solve $3(x - 2) = 2x + 5$.

① $3x - 6 = 2x + 5$

Expand first.

② $3x - 2x = 5 + 6$

Both terms change sign.

③ $x = 11$

④ Check: $3 \cdot 9 = 27$ and $22 + 5 = 27$ ✓

ANSWER

$x = 11$

EXAMPLE 2 Model answer, Q5: $a \parallel b$ and one angle is 115° .

① Vertical to it: 115° .

② Supplementary to it: 65° .

③ Corresponding and alternate: 115° .

④ Four angles are 115° and four are 65° .

ANSWER

Four of 115° , four of 65°

EXAMPLE 3 Model answer, Q8: $\{x + y = 9; x - y = 3\}$.

- 1 Add the equations: $2x = 12$.
Opposite coefficients of y .
- 2 $x = 6$
- 3 $6 + y = 9$, so $y = 3$.
- 4 Check both ✓ — the answer is the pair $(6; 3)$.

ANSWER

 $(6; 3)$ **03** Interactive model

Give back the diagnosis table with each script and ask every pupil to name their own two weakest blocks; they arrive in Grade 8 knowing where to look.

The year in twelve questions

$-7 + 12 - (-5)$ equals:

0

10

-10

24

$\frac{2}{3} - \frac{1}{4}$ equals:

$\frac{1}{12}$

$\frac{5}{12}$

$\frac{1}{7}$

$\frac{3}{4}$

$(3x - 2)^2$ equals:

$9x^2 + 4$

$9x^2 - 12x + 4$

$3x^2 - 12x + 4$

$9x^2 - 6x + 4$

$x^2 - 49$ factorises to:

$x^2 - 9$ $x + 3$ equals:

$3(x - 2) = 2x + 5$ gives:

On parallel lines, an angle of 115° has co-interior angle:

Angles in the ratio $2 : 3 : 4$ are:

$40^\circ, 60^\circ, 80^\circ$

$30^\circ, 60^\circ, 90^\circ$

$20^\circ, 30^\circ, 40^\circ$

$45^\circ, 60^\circ, 75^\circ$

$y = 2x - 4$ cuts Ox at:

$(0; -4)$

$(2; 0)$

$(4; 0)$

$(-2; 0)$

$\{ x + y = 9 ; x - y = 3 \}$ gives:

$(6; 3)$

$(3; 6)$

$(9; 3)$

$(4; 5)$

3-figure codes from 10 digits, all different:

1000

720

120

30

Handshakes among 10 people:

45

90

100

20

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Annual control work	Yillik nazorat ishi	Годовая контрольная работа
Revision	Takrorlash	Повторение
Method	Usul	Метод
Accuracy	Aniqlik	Точность
Diagnosis	Tashxis	Диагностика
Preview	Oldindan tanishtirish	Предварительный обзор
Progress	Rivojlanish	Прогресс
Grade 8	8-sinf	8 класс

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 tests:

integers and fractions

equations

geometry

counting

Q2 tests:

fractions

the abridged formulae

systems

probability

Q3 requires:

a graph

an excluded value

a construction

a ratio

Q5 tests:

triangles

angles on parallel lines

constructions

coordinates

Q8 must be answered with:

one number

an ordered pair

a graph

an equation

Q9 uses:

the addition rule

the multiplication rule without repetition

a ratio

a proportion

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $-7 + 12 - (-5)$

ANSWER 10

2. $\frac{2}{3} - \frac{1}{4}$

ANSWER $\frac{5}{12}$

3. $(3x - 2)^2$

ANSWER $9x^2 - 12x + 4$

4. $x^2 - 49$

ANSWER $(x - 7)(x + 7)$

5. $\frac{x^2 - 9}{x + 3}$

ANSWER $x - 3$

6. $3(x - 2) = 2x + 5$

ANSWER $x = 11$

7. Angles in the ratio 2 : 3 : 4

ANSWER $40^\circ, 60^\circ, 80^\circ$

Medium · the standard question

7 PROBLEMS

1. On parallel lines, the co-interior angle of 115°

ANSWER 65°

2. $y = 2x - 4$: both intercepts

ANSWER $(0; -4)$ and $(2; 0)$

3. $\{x + y = 9; x - y = 3\}$

ANSWER $(6; 3)$

4. 3-figure codes, all digits different

ANSWER 720

5. Handshakes among 10 people

ANSWER 45

6. A cuboid $5 \times 4 \times 3$: its volume

ANSWER 60 cm^3

7. Share 120 in $2 : 3$

ANSWER 48 and 72

Hard · stretch and reason

7 PROBLEMS

1. Factorise $2x^2 - 18$ completely

ANSWER $2(x - 3)(x + 3)$

2. Simplify $\frac{1}{x - 1} - \frac{1}{x + 1}$

ANSWER $\frac{2}{x^2 - 1}$

3. Solve $\{ 2x + 3y = 12 ; x - y = 1 \}$

ANSWER $(3; 2)$

4. An exterior angle of 125° , one remote angle 47°

ANSWER 78°

5. A right triangle with hypotenuse 16 and an angle of 30°

ANSWER The opposite leg is 8

6. How many 3-figure codes contain at least one 7?

ANSWER 271

7. Two numbers add to 26 and differ by 8

ANSWER 17 and 9

07 Homework

Over the summer — 5 tasks

Two blocks, chosen from your own diagnosis, are worth more than all eight done lightly.

1. Name your two weakest blocks from the diagnosis table.
2. Rework every question of the annual paper you lost a mark on.
3. Write out the five abridged multiplication formulae from memory and check them.
4. Solve ten linear equations and ten systems from the textbook exercises.

5. Read ahead: find out what an irrational number is, and bring one example to the first Grade 8 lesson.